

Online monitoring of socially compliant autonomous driving

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Abstract

Autonomous vehicles are assessed for collision safety, yet crash-free interactions can depart from human behaviour. We monitor conditional social atypicality by comparing a published trajectory-derived interaction-preference reading with a situation-conditioned human range, withholding judgement when the reading or human support is unavailable. Human interactions exhibit complementary roles; conditioning narrows the reference by about one-fifth while maintaining near-nominal coverage. In matched scenarios, automated systems are flagged about twice as often as human drivers, with the larger excess on the accommodating side. Assertive-side flags accompany tighter routine interactions for both vehicles without a detected increase in emergency-tail rates. A 40-participant evaluation of selected segments finds higher counterpart-position atypicality ratings for both outside-range categories than for within-range segments. Comfort and reported interaction cost distinguish their directions, with larger differences on the assertive side. The framework connects departures to subjective experience and vehicle-motion consequences while keeping social appropriateness and realised harm as distinct questions.

1 Introduction

Autonomous vehicles (AVs) must share the road with human drivers, cyclists and pedestrians whose behaviour is shaped not only by physical constraints but also by implicit social norms [1–3]. A vehicle can be *collision-safe* yet not *interaction-benign*: a manoeuvre that is collision-free in isolation may still be aggressive, hesitant or norm-violating in its interaction context, eroding the predictability on which mixed human–machine traffic depends [4–6]. Machine behaviour in human collectives is now an object of empirical study in its own right [7]: controlled experiments show that machine agents embedded in human groups reshape coordination and cooperation [8–10], and can suppress beneficial social norms such as reciprocity, with collective consequences [11]. Conventional safety verification asks “will this trajectory collide”; a socially aware AV also faces a different question: “is this way of interacting still understandable and acceptable to the humans around it?”

Two established families of methods address adjacent but distinct questions. Offline behavioural evaluation asks whether an AV, on average, drove like a human, typically by reporting a single aggregate sociality score over a dataset [12–15], and no test-mileage budget can in any case certify rare interactive behaviour by aggregate demonstration [16, 17]. Formal safety verification asks whether a trajectory will collide or violate a hard constraint, and can guarantee a formally specified notion of legal safety — within the assumptions of its model — as an online layer around a planner [18–21], an assurance now being codified in standards [22, 23]. Neither answers the moment-to-moment question: given the current interaction state, is this behaviour within the range that human drivers exhibit in the same situation? That question is local in time, conditional on context, and graded rather than binary.

We make autonomous-driving social deviation *visible online* by comparing current interaction behaviour with a human reference range, while separating moments that support a judgement from those that do not. The interpretation of a departure then requires evidence beyond interval membership. Do people also perceive the behaviour as atypical? Do more assertive and more accommodating departures correspond to the same comfort

and interaction cost? We address these questions through two complementary dimensions: subjective experience of selected interactions and vehicle-motion consequences following flagged moments. Neither dimension is treated as a necessary intermediate step to the other; direct judgements of appropriateness and evidence of realised harm remain separate validation targets.

The monitored quantity is the interaction preference value (IPV), a trajectory-derived measure of how an agent balances its own progress against the group's. The trajectory-based estimator was introduced and validated in controlled VGIM interactions in our previous work [24]; here we use its frozen online implementation as the behavioural reading supplied to the human-reference monitor. A related inverse-game formulation provides complementary evidence for quantifying social patterns from trajectories [25].

Three distinctions bound the claims. First, *socially atypical is not socially inappropriate*: lying outside the human range is a statement about typicality. Subjective experience and vehicle-motion outcomes help interpret that departure without converting it into a normative verdict. Second, *collision-safe is not interaction-benign*. Third, *not readable is not neutral*: before an interaction is behaviourally informative or after it has resolved, the IPV may be weakly identifiable or not applicable, and this must not be read as compliant. A responsible social monitor must know when *not* to judge; abstention is an explicit boundary of responsible judgement, not a failure mode, in the tradition of prediction with a reject option [26–28]. The human range is, moreover, a *reference range* in the sense familiar from clinical medicine and normative modelling [29–31], not a demand to imitate one human trajectory.

This framing differs from three existing paradigms. Unlike offline AV evaluation [12, 13], it is online, per-timestep and context-conditioned. Unlike runtime verification of formal specifications [32–35], the monitored property is an empirical reference distribution estimated from human data. Unlike socially compliant planning [36–40], it does not encode sociality into a planner's objective but provides a pluggable, planner-agnostic monitor that any planner can be checked against. Closest to our aim, online monitors exist for codified legal rules [41]; unusual driving can be flagged as unexplained deviation

from learned traffic behaviour [42] or as violated expectation [43, 44]; and AV conduct is benchmarked offline against reference models of a competent human driver [45, 46]. None of these reads a social preference, refers it to a calibrated situation-conditioned human range, or separates the moments at which a social judgement is meaningful from those at which it is not.

This work reframes social compliance as the online monitoring of conditional social atypicality. Human interaction is relational and context-dependent: partners take complementary roles, yet the reference needed online can be determined from the current observable situation without reading a hidden intention. Built on that situation-conditioned range, the monitor is calibrated and auditable at runtime, and it abstains whenever the interaction is unreadable or the situation lacks comparable human support. We then apply the unchanged instrument to a matched-scenario real-vehicle benchmark, where 19 automated systems and 20 licensed drivers face the same 15 interactions. Automated systems are flagged about twice as often as the matched human drivers, with the larger excess on the accommodating side. At assertive-side flagged moments, ordinary interaction quality tightens for both vehicles without a corresponding increase in emergency-tail events. The human arm checks alarm inflation after transfer to a new country and apparatus and examines the same consequence signature in human-driven interactions. A separate subjective evaluation with 40 participants asks how 90 selected interaction segments are experienced from the ego and counterpart positions. It links both outside-range categories to perceived atypicality from the counterpart position and identifies larger comfort and reported-cost differences on the assertive side. Together, the studies separate how frequently a departure occurs, its direction, and its experiential and motion-related meaning.

2 Results

2.1 A human-reference monitor makes conditional social atypicality visible online

The trajectory-based IPV estimator was previously validated in controlled VGIM interactions [24]; here it is held fixed and used as the frame-level behavioural reading supplied to the human-reference monitor. Figure 1 summarises the monitoring pipeline, including the trajectory-derived reading, the context-conditioned reference, explicit abstention and one unchanged interaction example.

2.2 Human interaction runs on a two-sided script, and the reference must condition on the situation

We next ask what a reading should be compared against, beginning with the structure of human interaction in the space that the reading describes. We analysed 38,228 interaction cases from four naturalistic sources within the InterHub corpus [47] (Waymo 23,218; nuPlan 7,499; Lyft 5,105; Argoverse-2 2,406; 3,695,981 trajectory frames), summarising each agent by the mean IPV over its valid interaction segment. Seen whole, an interaction between two humans follows a two-sided script rather than two independent scores. Plotting each valid case as the pair of its partners' summaries concentrates the mass where one partner sits below zero while the other sits above it. Across the 34,757 cases in strata that support matched re-pairing, the within-interaction correlation is -0.338 (95% CI $[-0.351, -0.325]$), whereas re-pairing partners across cases within matched scenario strata gives $+0.041$ ($[+0.030, +0.052]$); an independently built implementation returns -0.339 , and the correlation is negative in every source (Fig. 2a,b).

The division of roles is legible early. Readings from the first 5% of an interaction identify which partner has the higher mean IPV in the final quarter at AUC 0.75, rising to 0.81 by half-way, while early kinematics over the same windows remains near 0.56–0.57 (Fig. 2c). The target is the late-interaction IPV role, not observed passage

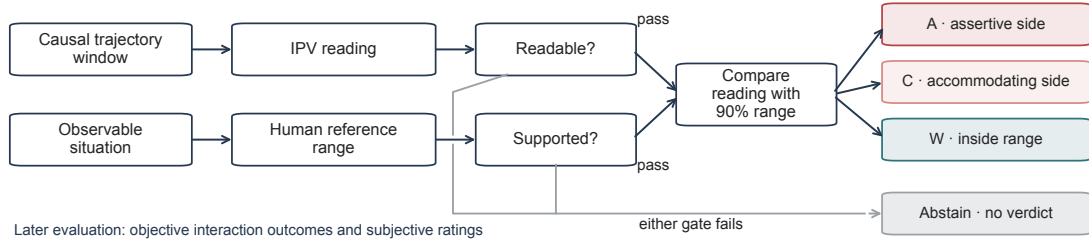
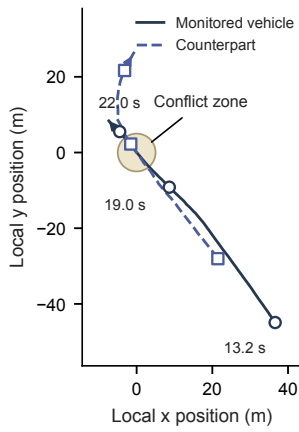
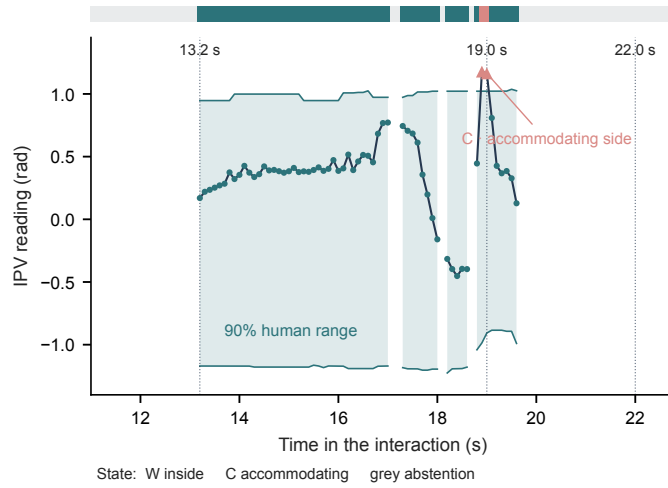
a Two inputs, two evidence gates, one comparison**b One observed interaction****c Reading and dynamic 90% range**

Figure 1. A trajectory-derived reading is compared with a situation-conditioned human range, with explicit abstention. **a** A causal trajectory window supplies the IPV reading; the current observable situation supplies its human reference range. Readability and human support are checked before a verdict is issued. Outside-range directions are more assertive (A) and more accommodating (C); W is within range. Objective outcomes and subjective ratings evaluate departures and do not enter the monitor. **b** One illustrative interaction, with ego and counterpart trajectories and time markers. The conflict-zone circle is a schematic locator, not a measured road boundary. **c** Its frame-level reading and frozen 90% reference, with a verdict strip. Grey intervals and gaps denote abstention, never a zero reading or a within-range verdict. No curve is interpolated across those gaps. The example is descriptive and contains within-range and accommodating verdicts. Estimator mechanics and reference-population summaries appear in Supplementary Fig. S4 and Extended Data Fig. ED4. Source Data accompany this figure.

order. The script is also keyed to the situation before the conflict is joined: turning vehicles read $+0.079$ rad further toward giving way than their straight-driving partners ($[+0.063, +0.095]$, $n = 3,565$ pairs), in the same direction in all four sources, with one source's interval admitting zero (Fig. 2d). Social behaviour at this level is relational—organised by partner, role, geometry and risk—rather than a fixed disposition of an individual.

No single number attached to a role reproduces that script. What holding right-of-

way contributes is not a fixed offset: across three bands of realised post-encroachment margin the priority minus non-priority difference is $+0.058$ ($[+0.050, +0.067]$, $n = 9,799$), $+0.001$ ($n = 22,833$) and -0.034 ($[-0.045, -0.023]$, $n = 4,863$; Fig. 2e). These offline bands are used only to describe the human data and are never read by the online monitor. A coarse geometric prior points in the same direction across all four sources (Fig. 2f), and the dependence is preserved under dropping the largest source and alternative geometry and risk binnings.

The human reading distribution and the situation-dependent reference provide complementary views of this heterogeneity (Extended Data Fig. ED4). The distribution contains 486,660 readable natural-driving test moments. Separately, across the automated benchmark’s 67,861 candidate moments, the predicted 90% reference width spans 1.05–2.37 rad. Among three illustrative deployment classes, the interval formed by the median lower and upper bounds is narrowest for crossing with priority and widest for merging with equal priority. These are summaries over deployment candidates, before gate filtering; the actual range varies within each class and the two median endpoints need not belong to the same moment. We report this as a regularity of the observed corpora rather than a transferable law: fitting the full state description on every source but one explains at most $+0.03$ of the variance in the held-out source and falls below zero in two (Extended Data Fig. ED3a; Methods 4.4).

2.3 A situation-conditioned human range is a calibrated monitor

Establishing that human behaviour is context-dependent does not by itself say what an online monitor must read. We define online monitoring as a membership test: the ego’s current IPV should lie within a human reference range determined by the current observable situation,

$$\begin{aligned} \hat{\theta}_i(t) &\stackrel{?}{\in} I_{\text{human}}(z_t), \\ z_t &= (\text{geometry, role, progress, online-computable risk proxies,} \\ &\quad \text{relative kinematics, support}), \end{aligned} \tag{1}$$

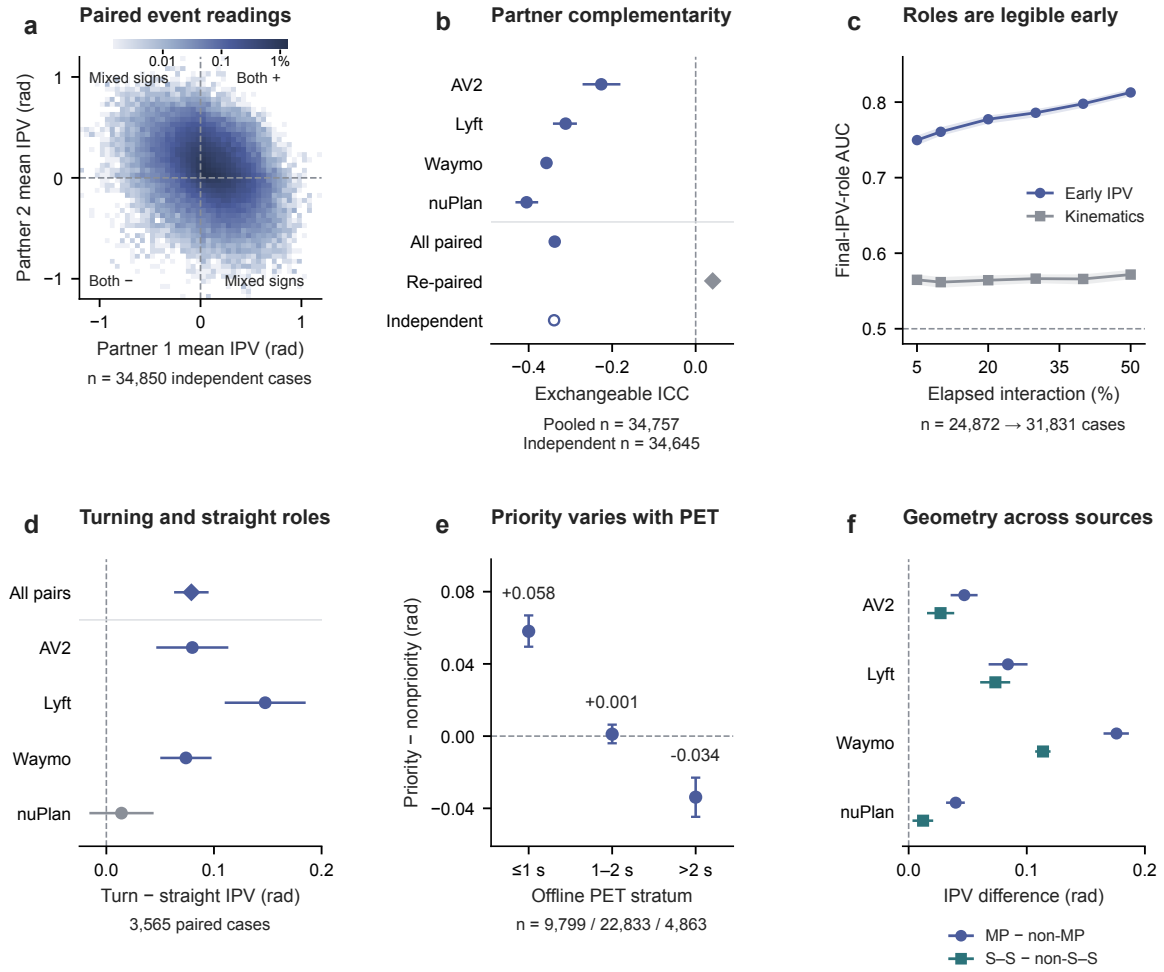


Figure 2. Human interaction is relational and context-dependent. **a** Joint distribution of event-level partner IPV summaries for 34,850 independent valid cases. Each case is displayed in both partner orders; this symmetry does not double n . **b** Exchangeable intraclass correlations (ICCs) and matched re-pairing references, by source and pooled. Source estimates use 34,850 cases in total, while the matched-support pooled analysis includes 34,757 cases; its correlation is -0.338 versus $+0.041$ after re-pairing. The independent implementation uses 34,645 cases and has no supplied interval. **c** Existing out-of-fold AUC trajectories for predicting which partner has the higher late-interaction IPV, defined from the final quarter. This label is not observed passage order. **d** Paired turning-minus-straight IPV differences, pooled and by source, retaining the source whose interval includes zero. The pooled difference is $+0.079$ [$+0.063, +0.095$] rad across 3,565 pairs. **e** Priority-minus-non-priority differences across three realised post-encroachment-time (PET) bands. PET is an offline descriptive stratification, never an online input. **f** Merge/pass and same-direction geometry contrasts by source. Intervals are the existing 95% source intervals: case bootstrap for source correlations, scene-cluster bootstrap for the pooled ICC and AUC, central matched-reshuffling intervals for the null, paired-difference t intervals for turning contrasts, case bootstrap for PET strata, and two-group bootstrap for geometry. Cross-source prediction and reference coverage are retained in Extended Data Fig. ED3. Source Data specify each panel’s population and interval definition.

evaluated only where the interaction is active and the ego is readable; the risk proxies are those computable from information available at time t (no observed post-encroachment measure enters online). The range is the conditional distribution of the human current IPV given z_t , calibrated by split-conformal prediction [48, 49] and abstaining when the state is unsupported. Conditioned on the current situation, the human range is sharper than a single global range—about a fifth narrower at the 80% and 90% levels (21% and 20%), at coverage within 0.6 percentage points of nominal—and it stays informative at the strictest level, where the global range spans essentially the whole admissible scale and retains almost no power to discriminate. In absolute terms the conditioned 90% range still averages 1.87 rad (5th–95th percentiles 1.35–2.28 rad) of the 2.36 rad candidate span: the reference is wide, and a flag therefore marks a clear departure from the human range rather than a fine discrimination. The coverage statement is empirical and marginal over accepted moments; it is not a per-situation or per-interaction guarantee (Methods 4.4). Where a reading is available, the monitor abstains for lack of human support on only a small fraction of readable moments (5.08% on the held-out natural-driving test fold; abstention on the real-vehicle benchmark is larger and is reported with its reasons in Section 2.4 and Methods), giving an auditable runtime monitor of conditional social atypicality (Fig. 3). The range is not a fixed corridor: it rises, falls and changes width as the situation changes, so that the same reading can sit inside it in one situation and outside it in another (Fig. ED1).

A striking feature organises this result. Although the interacting partners’ IPVs are correlated at the event level (above), *adding the counterpart’s own inferred preference to the situation does not measurably sharpen the reference range*; conditioning on the ego agent’s own behavioural history does not either. Both are retained only as ablations. Thus the event-level complementarity does not translate into measurable additional sharpening from the counterpart-IPV channel in this implementation. The current observable situation supplies the deployed reference without requiring that additional inferred preference. The monitor returns an explicit abstention when the situation lacks human support or the reading is unavailable. Its supported interpretation remains conditional social *atypicality*.

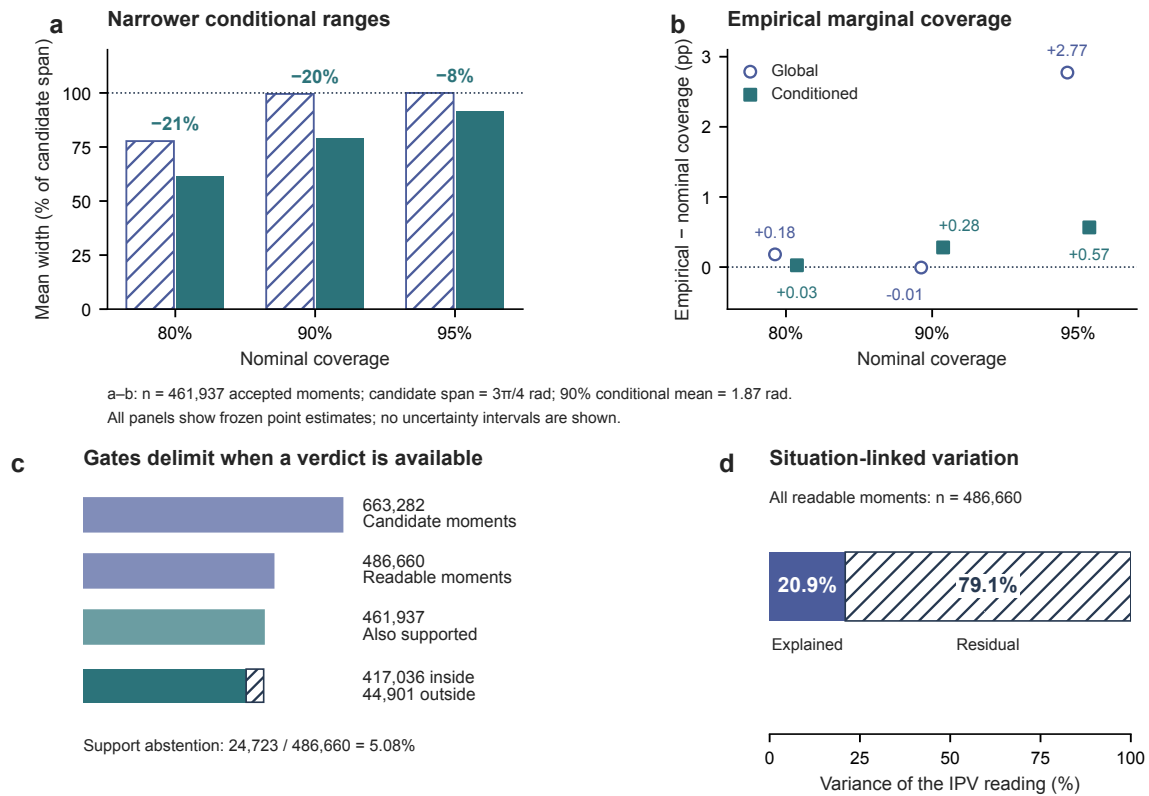


Figure 3. Conditioning sharpens the human reference at near-nominal coverage, while the gates delimit its use. **a** Global and situation-conditioned interval widths at nominal 80%, 90% and 95% coverage, expressed as a percentage of the seven-candidate span $3\pi/4 \approx 2.36$ rad. Reductions of 21%, 20% and 8% are calculated from unrounded widths; the conditional 90% width averages 1.87 rad. **b** Empirical coverage minus nominal coverage in percentage points, with zero marking nominal coverage. Both panels use the same 461,937 accepted moments and describe marginal coverage, not a guarantee for each situation. **c** Nested natural-driving test-fold counts: 663,282 candidate moments, 486,660 readable, and 461,937 also supported by the human reference; 417,036 are within the 90% range and 44,901 outside. Human-support abstention is 24,723/486,660 = 5.08% of readable moments. Failing either gate returns no verdict. **d** The situation explains 20.9% of reading variance across all 486,660 readable moments, using $1 - \text{SSE}/\text{SST}$ for the conditional median against the test-fold mean. The remainder is unexplained by this predictor and is not assigned a psychological interpretation. Source Data accompany this figure.

2.4 From atypicality to interaction consequences

We ask what an atypicality verdict corresponds to in the interaction that follows, when the same situations are driven by different automated systems. We pursue this on a matched-scenario real-vehicle benchmark [50], in which identical scenarios are replayed against many independent driving systems.

The monitor is, first, able to speak. The benchmark replays 15 scenario templates against 19 independent driving systems (267 scenario runs). Across its 67,861 candidate moments the monitor returns a verdict on 14,099 (20.8%), with no engineering failures in this replay — no solver or pipeline failure on any candidate moment — and in 231 of the 267 scenario runs (86.5%) it speaks at least once. At the moment level, 55.3% of candidate moments carry discriminative information between the candidates, and 20.8% additionally lie within human support; at the run level, none of the 36 runs in which the monitor never speaks is silenced by readability alone — in every such run there are readable moments, and what is missing there is comparable human behaviour for the situation. Abstention here is a statement about the reference rather than about the vehicle, and it is auditable as such.

If human interaction follows a two-sided script (Section 2.2), a departure by one partner should be visible both in that vehicle’s own margin and in what the counterpart absorbs. Where the monitor does speak, atypicality on the assertive side—a current preference that weights the counterpart’s cost less than humans do in the same situation—is followed by a measurably tighter interaction for *both vehicles*. The ego’s own margin to its counterpart contracts at the comfortable end of its distribution: the upper quartile falls by two-fifths, while the lower quartile—the end where a margin would actually be pressing—is not compressed at all. The counterpart, read independently from the other vehicle’s own control record, carries nearly twice the routine speed variation over the same window, and roughly twice the routine speed reduction (+2.6 and +1.4 km h⁻¹ as differences, which do not depend on how small the routine value is). The two sides are measured from different sources and different quantities, so their agreement is a convergence rather than a restatement (Fig. 4); the association survives a placebo test

that reassigns whole flag sequences across scenario runs (Methods 4.5); a single run containing flagged assertive moments is shown for illustration in Fig. ED2.

The same outcome measures, applied unchanged to the accommodating side, do not reproduce that signature: neither the counterpart’s speed variation nor its speed reduction separates detectably from within-range moments, and neither the median nor the upper quartile of the ego’s own margin does either. The two sides are also compared with each other directly, rather than each against the within-range group in turn, and on the quartile where the assertive side compresses they differ (Methods 4.5). The result is direction-specific: assertive-side departures accompany compression of the measured routine vehicle-motion outcomes, whereas accommodating-side departures show no corresponding compression in this battery. The latter finding concerns these objective endpoints; subjective experience is examined separately in Section 2.6.

Counting both sides, 9.8% of judgeable moments (candidate moments that pass both gates; Methods 4.4) are flagged (1,388 of 14,099; 519 on the assertive side and 869 on the accommodating side). These fractions describe the frequency and direction of departures. The outcome analyses describe what follows flagged moments. The next section places the flag rate alongside a matched human control.

This is a compression of ordinary interaction quality, not a shift of mass into the tail. At the emergency thresholds—a closing margin short enough to demand immediate action, or braking hard enough to count as evasive—assertive-side moments are roughly half as frequent as typical ones (0.46 to 0.61 times) at every threshold whose interval excludes a zero difference (margins under 1, 1.5 and 2 s; braking beyond -2 , -3 and -4 ms^{-2}); at the widest margin tested (under 3 s) the interval admits no difference. On the accommodating side the only threshold whose interval excludes zero is the tightest margin, and it points the same way, with fewer short margins rather than more. The lower quartile of the ego’s margin is not compressed on either side, sitting marginally wider for flagged moments than for typical ones. What the monitor registers is that ordinarily comfortable margins tighten, not that dangerous events become more common. The accommodating side shows no corresponding compression of these routine vehicle-motion

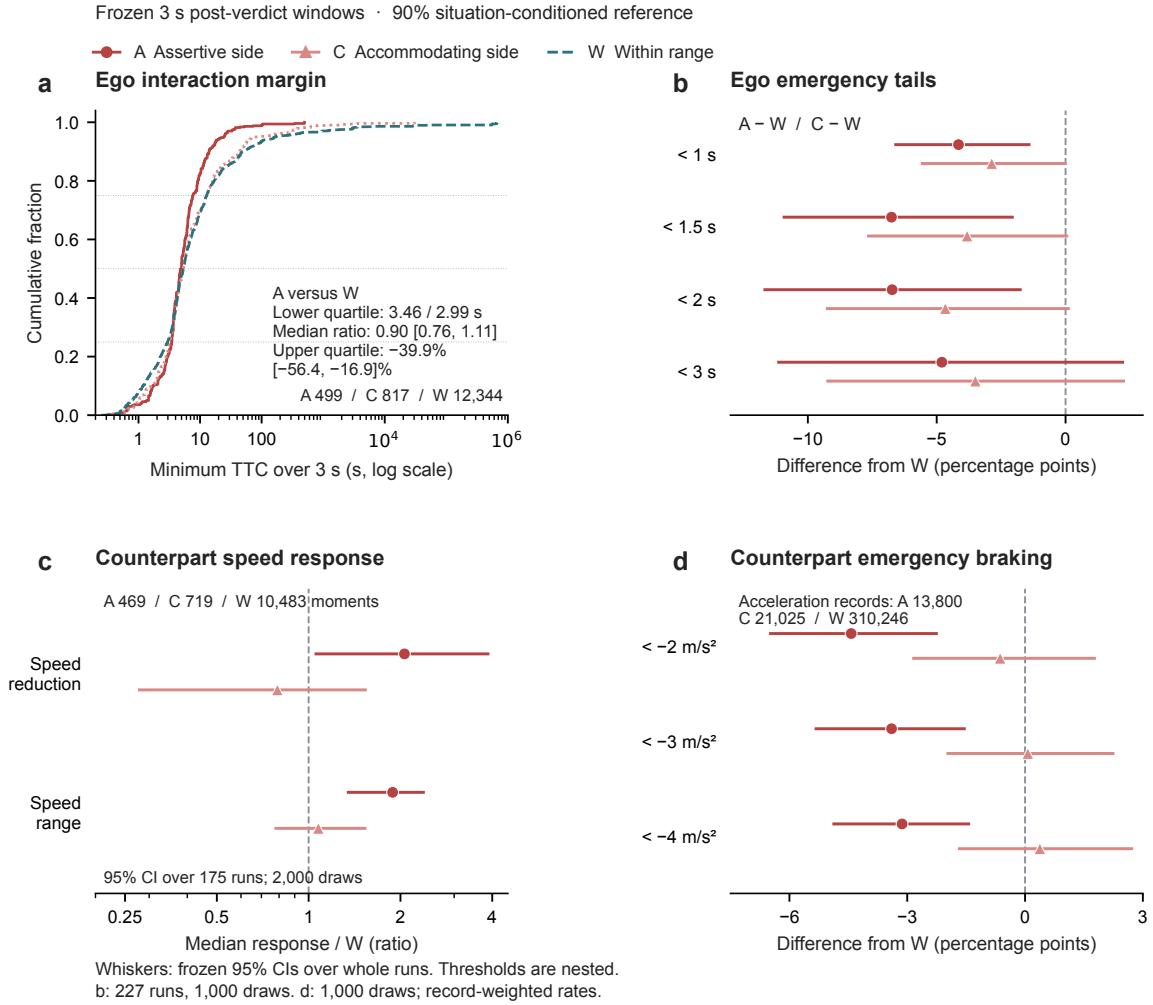


Figure 4. Assertive-side departures accompany tighter routine interactions without a detected increase in the tested emergency tails. A denotes the assertive side, C the accommodating side and W within range. **a** Cumulative distributions of minimum ego-to-counterpart time-to-collision over the frozen three-second post-verdict window, omitting non-closing frames; the full observed range is shown on a logarithmic axis. The assertive upper quartile is 39.9% lower [16.9, 56.4], while the lower quartile is not compressed and the median contrast remains unresolved. **b** A-W and C-W share differences for all four nested ego-margin thresholds, in percentage points. **c** Counterpart speed-reduction and speed-range median ratios relative to W, for both directions. Existing ratio intervals are retained because corresponding accommodating-side absolute-difference intervals are unavailable; absolute differences are not inferred from them. **d** Both direction-specific share differences for all three nested counterpart-braking thresholds, using acceleration records. Panels **a,b** use 499 A, 817 C and 12,344 W moments across 227 scenario runs. Counterpart data use 469 A, 719 C and 10,483 W anchors across 175 runs; **d** pools 13,800, 21,025 and 310,246 acceleration records, respectively. Intervals are pointwise 95% whole-run bootstrap intervals: 2,000 draws for quantile and speed ratios and 1,000 for threshold differences. Moments and records are not independent resampling units. Full steering and threshold results remain in Source Data. All associations are descriptive.

outcomes.

Compression of interaction quality is not yet a harm claim, and we do not make one. Under the present definitions of a realised interaction failure, atypicality does not robustly or specifically predict harm outcomes; the binding limitation is upstream, in that identifying and segmenting interaction-failure events is itself not yet reliable. The present evidence therefore bounds the harm relationship rather than confirming it.

What, then, does a social flag register that a conventional safety check does not? The routine interaction differences above occur without an observed rise in the tested emergency-tail rates. Thus the social reading and the threshold-based safety endpoints describe different aspects of these interactions; the routine differences are not evidence of more emergency events.

2.5 The reference is audited by the population that defined it

The reference range was learned from natural human driving recorded in other countries on open roads, and every verdict above rests on the premise that this reference remains readable in a new country, on a closed course, in deliberately staged conflicts. We test that premise directly: licensed human drivers drove the same 15 scenarios in a real vehicle, against the same counterparts under the same control, and the unchanged monitor—same estimator, same gates, same frozen reference—audits them exactly as it audits the automated systems.

Three observations follow (Fig. 5). First, the instrument survives the move. The population that defined the reference lies outside it on 9.72% of held-out judgeable moments at the 90% level—the instrument’s native alarm rate. Human drivers on the course are flagged on 5.0% of judgeable moments (786/15,598), no more often than that native rate: carrying the reference across country, apparatus and task does not inflate its alarms. The course human rate therefore serves as the matched calibration control; the automated systems are evaluated against the same frozen reference in the same scenarios, and their corresponding rate is reported below. The two settings are not, however, interchangeable. Drivers on the course were asked to drive as they normally would, but the scenario set

is not ordinary driving: these are staged conflicts selected for a benchmark, and they are harder than what a naturalistic corpus mostly contains. The cross-corpus reading is therefore an alarm-inflation check—does the instrument fire more often once it is carried somewhere new—and not a comparison of levels. The comparison of levels is the one made inside the course, where both kinds of driver meet the same fifteen scenarios and the same counterparts under the same control. How often the monitor fires at all is a property of the scenarios it is pointed at: across these fifteen the human flag rate runs from 1.7% to 12.4% and the automated rate from 2.7% to 17.4%, so an absolute alarm rate is not a portable quantity. That is why the audit carries its own human arm rather than treating the corpus rate as a yardstick. The yardstick itself never changes: both arms are judged against the same frozen reference, built from human–human interaction alone, and the question asked of each is whether the behaviour falls inside the human range for this situation—never whether it resembles the other arm. Second, the flag means the same thing for either kind of driver. At the moments where a human is flagged on the assertive side, the same consequence signature appears that Fig. 4 documents for automated vehicles: the ego margin’s upper quartile contracts by about three-tenths, the counterpart absorbs 2.5 times the routine speed reduction, and every emergency tail is rarer, not more common. The verdict marks a property of the interaction, not a property of machines. Third, under this common instrument the automated systems are flagged 2.0 times as often as the matched human drivers (1.95, 95% CI [1.62, 2.39] resampling scenarios; 9.8% vs 5.0% of judgeable moments), and the per-scenario flag rate is higher for the automated systems in 15 of 15 scenarios—in every resampled scenario set the ratio stays above parity. The population-level rates answer a calibration question; the moment-level signature answers what a flag means; neither is a ranking of vehicles or drivers.

The side-specific split sharpens the comparison. Automated systems exceed matched humans primarily on the accommodating side (6.16% versus 2.25%), whereas the assertive-side difference is smaller (3.68% versus 2.79%). The larger machine–human excess therefore lies on the side for which the frozen vehicle-motion battery detects no

corresponding compression signature. Whether that side also differs in human experience is a separate question, addressed next.

2.6 Subjective experience distinguishes the two directions of atypicality

We examined whether the monitoring categories were also reflected in human experience of selected interaction segments. This separate study comprised 40 participants in 20 pairs and 90 segments from 15 scenarios; 596 completed trials yielded 1,192 position-specific rating records. From the counterpart position, both assertive-side and accommodating-side segments were rated as more atypical than within-range comparison segments: mean within-participant differences were $+0.639$ [$+0.318, +0.992$] and $+0.6825$ [$+0.548, +0.817$] points, respectively, on a 1–7 scale. These contrasts survived Holm correction across all 30 exploratory comparisons. Thus the supplied monitoring categories have a perceptual counterpart in this selected stimulus set (Fig. 6; Supplementary Table S2).

Comfort and reported interaction cost distinguished the two directions more clearly. Relative to within-range segments, assertive-side segments received counterpart-position comfort ratings 1.091 points lower and interaction-cost ratings 2.070 points higher. Direct comparison with accommodating-side segments gave a comfort difference of -0.870 [$-0.974, -0.766$] points and a reported-cost difference of $+1.163$ [$+1.019, +1.300$] points. Ego-position comfort followed the same ordering, with the assertive category below both other categories. The result therefore concerns which side of the reference is crossed as well as whether a segment is outside it.

Accommodating-side segments also differed from within-range segments: counterpart-position comfort was 0.221 points lower ($[-0.351, -0.100]$) and reported interaction cost was 0.908 points higher ($[+0.771, +1.051]$). These smaller subjective differences complement the absence of a corresponding compression signature in the separate vehicle-motion analysis. They describe another dimension of interaction quality; the two studies do not establish a segment-matched dissociation or a causal effect of crossing a reference boundary.

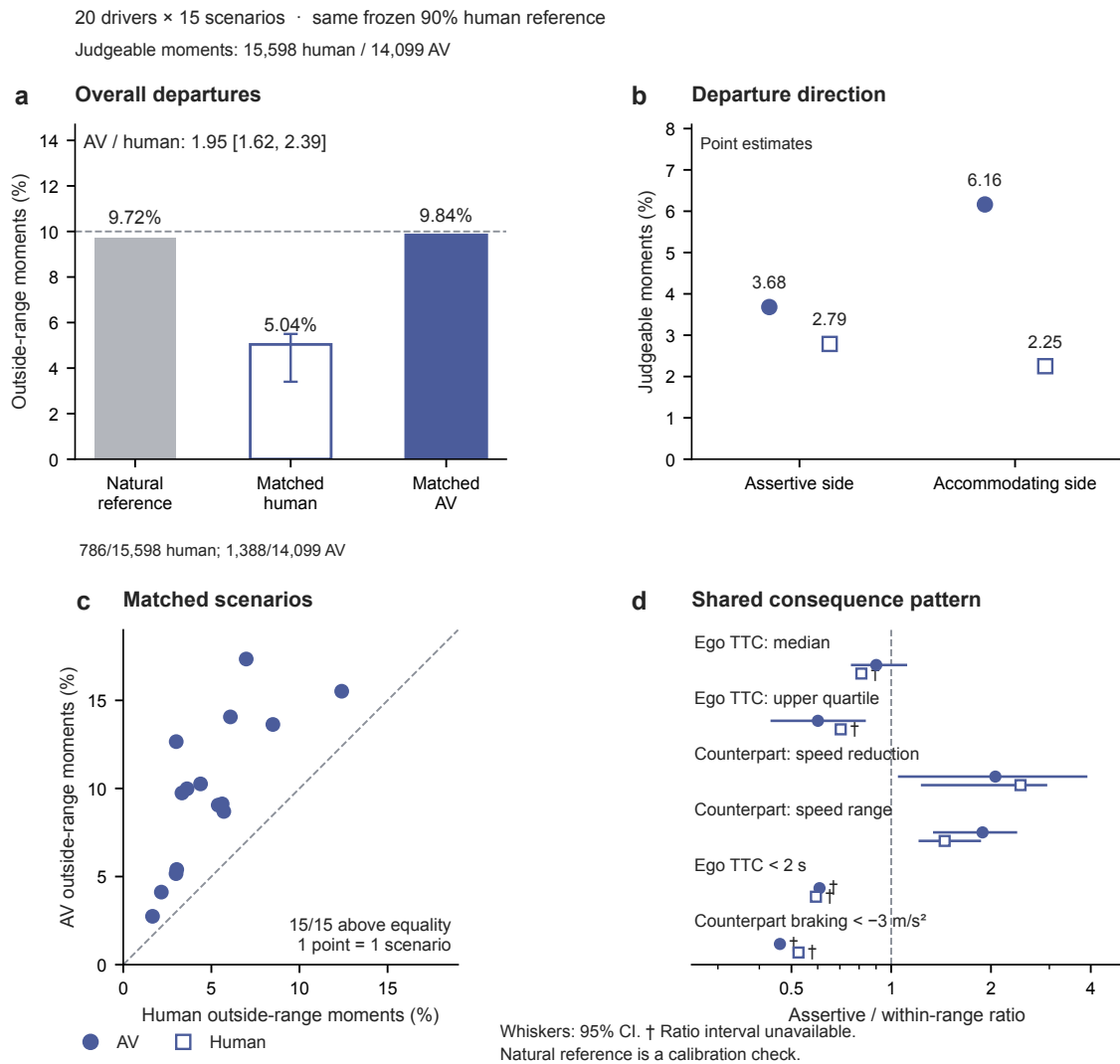


Figure 5. The same human reference audits matched human drivers and automated systems; their larger rate difference is accommodating. **a** Outside-range rates under the frozen 90% reference: the natural-driving test population, matched human arm and matched automated arm. The natural rate is an alarm-inflation check, not a same-condition comparison. The human rate has the existing driver-by-scenario bootstrap 95% interval of 3.4–5.5%; the dashed line marks the nominal 10% outside rate. The matched AV/human ratio is 1.95 [1.62, 2.39], from the existing scenario bootstrap. **b** Direction-specific point estimates: assertive AV 519/14,099 = 3.68% and human 435/15,598 = 2.79%; accommodating AV 869/14,099 = 6.16% and human 351/15,598 = 2.25%. Both denominators are judgeable moments in the corresponding arm; side-specific intervals are unavailable. **c** The 15 paired scenario flag rates with equal axes and the equality line, retaining all observed scenarios as point estimates. **d** The common assertive-to-within consequence signature, separating routine margins, counterpart response and emergency tails. Filled markers denote AVs and open markers humans. Existing ratio intervals are shown where available; daggers mark unavailable ratio intervals, including both human margin estimates and the emergency-tail ratios. Missing intervals do not establish precision or non-detection. Other nominal coverage levels are retained in Supplementary Fig. S3. Source Data document each endpoint’s denominator and interval.

The main comfort and cost directions were retained when categories were compared within shared scenarios. For assertive-versus-within comparisons, counterpart comfort was lower and reported cost higher in all eight shared scenarios; the corresponding assertive-versus-accommodating directions held in all 12 shared scenarios. These are scenario-composition sensitivity checks, not independent replications. Estimates and intervals preserve participant pairing and condition on the selected stimuli (Methods 4.7).

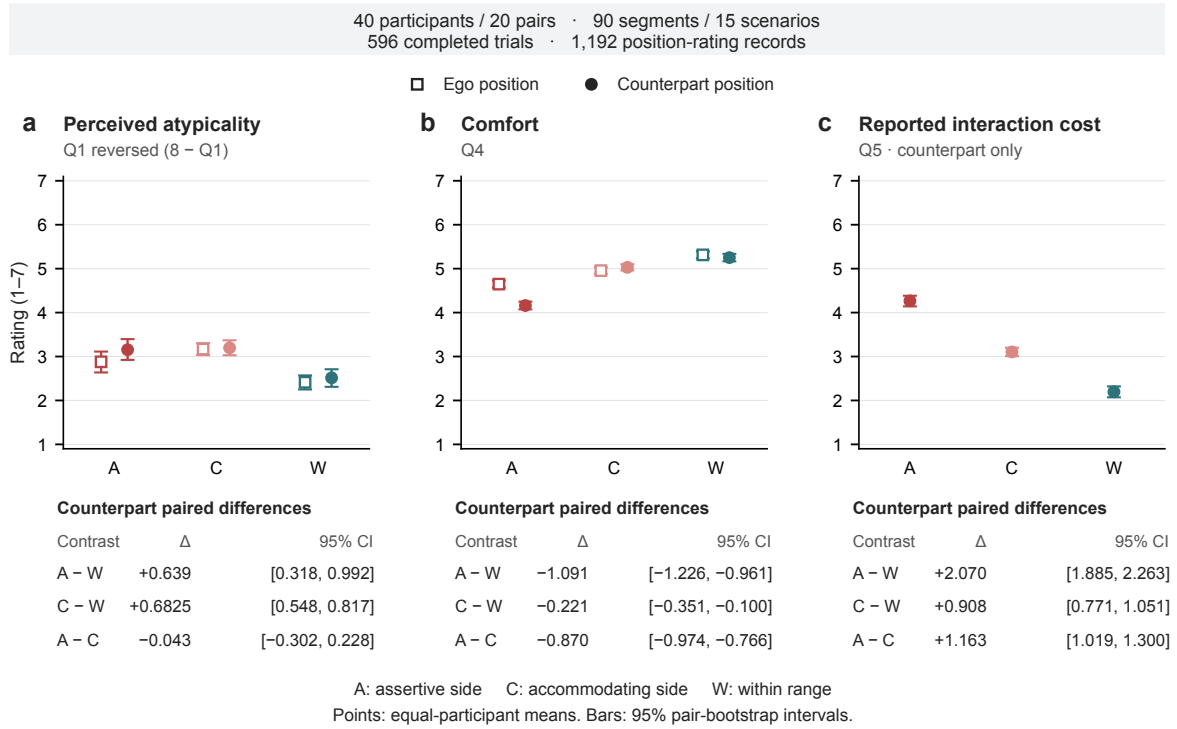


Figure 6. Subjective ratings distinguish perceived atypicality from comfort and reported interaction cost. **a** Perceived atypicality (8 - Q1), **b** comfort (Q4), and **c** reported counterpart interaction cost (Q5), for supplied assertive-side (A), accommodating-side (C) and within-range comparison (W) segments. Ego and counterpart positions are distinguished by marker shape and fill; Q5 is counterpart-only and structural ego blanks are not zeroes. Each position contributes 158 A, 239 C and 199 W ratings, from 40 participants in 20 pairs. Class means average repeated ratings within participant and then weight participants equally. Bars are existing pointwise 95% intervals from 10,000 participant-pair bootstrap draws, conditional on selected stimuli. Direct within-participant contrasts are displayed separately; significance is not inferred from overlap of class-mean intervals. The full 30-comparison family uses Holm correction (Supplementary Tables S2 and S3); ego A-W atypicality has adjusted $p = 0.0704$, and A-C non-detection does not imply equivalence. Scenario-composition sensitivity is shown in Supplementary Fig. S2. Segment labels do not specify every frame's verdict, and W may include abstention. Source Data contain aggregate results only.

3 Discussion

This work makes social monitoring an auditable question about behaviour in a situation: whether the current interaction lies within a human reference range, when that comparison is supportable, and what an observed departure means. Human behaviour is relational and context-dependent, with complementary roles that are legible early. The situation-conditioned reference is sharper than a global range at near-nominal coverage and withholds judgement when the reading or human support is missing. In matched real-vehicle scenarios, the reference does not inflate alarms for human drivers, and assertive-side flagged moments show a shared vehicle-motion signature in human and automated arms. The separate subjective study adds evidence that the selected outside-range segments are perceptibly atypical and that their direction matters for comfort and reported interaction cost.

Two distinctions organise these findings. First, interaction-level complementarity and the inputs needed by an online reference are different questions. Although partners' readings covary across interactions, adding counterpart IPV or ego self-history does not measurably sharpen the reference beyond the current situation in the evaluated implementation. The deployed monitor can therefore use the situation-conditioned range without those additional channels. This ablation result does not identify a psychological mechanism behind the partner relationship.

Second, the frequency of departures and their experiential meaning need to be reported separately. The larger automated-human excess occurs on the accommodating side, while the larger comfort and reported-cost differences in the subjective study occur on the assertive side. Accommodating-side segments nevertheless carry smaller subjective differences relative to within-range segments. Together, these observations motivate retaining direction when interpreting a flag rather than using its frequency as a proxy for interaction cost. The rates and ratings come from different analyses and are not combined into a population-level social-loss score.

Subjective experience and vehicle-motion consequences also provide complementary endpoints. The real-vehicle result is compression of routine interaction quality for the

ego and counterpart at assertive-side flagged moments, without a detected increase in the tested emergency tails. The absence of the same compression on the accommodating side applies to that fixed motion-based battery. Its lower comfort and higher reported-cost ratings in the subjective study show why a broader evaluation should retain an experiential dimension. The current studies do not resolve which individual motion features account for those ratings.

The empirical reference, subjective experience, direct normative judgement and realised harm remain distinct. The questionnaires support statements about perceived atypicality, comfort and the reported interaction-cost item; these endpoints do not establish psychological IPV recovery or a universal judgement of social appropriateness. The subjective analysis is exploratory and its participant-pair intervals condition on 90 selected segments. Within-scenario comparisons check composition sensitivity without establishing generalisation to all new stimuli. A reliable link to realised interaction failure remains unconfirmed under the present outcome definitions.

The human-driving arm carries a separate scope: participants drove an instrumented vehicle on a closed course against controlled counterparts, supervised and instructed to drive naturally, with a fixed scenario sequence. Its protocol and consent cannot be substituted for those of the subjective study. The evidence covers pairwise vehicle-vehicle interactions at mapped conflict points; cyclists, pedestrians and multi-agent scenes remain outside the present reference. Beyond driving, the framework offers a template for comparing autonomous-agent behaviour with human reference ranges while separately examining experience and interaction outcomes and explicitly withholding unsupported judgements [7].

4 Methods

Note on scope. The main text states the social-monitoring argument; this section records the estimator, the reference-range model and its ablations, the calibration and abstention machinery, the real-vehicle consequence protocol, the matched human-driving arm and

the separate subjective-evaluation analysis. Throughout, *online* means that every input to a verdict is available at decision time (non-anticipation verified); end-to-end real-time operation on vehicle hardware is not evaluated in this paper.

4.1 Online likelihood-based inverse-planning IPV estimation

The trajectory-based IPV estimator was introduced and validated in controlled VGIM interactions, where imposed interaction-preference patterns were identified from trajectories and improved interactive-motion prediction [24]. We adopt the published estimator as a fixed component of the monitor; Supplementary Note 1 summarises the earlier validation and the present online implementation.

For a grid of candidate preferences $\{\theta_k\}$, candidate trajectories $\tilde{X}_a(W_t; \theta_k)$ are generated and scored against the observed trajectory, the counterpart trajectory and the static map-lane-centreline reference. The frame-level reading is the likelihood-weighted mean,

$$\begin{aligned} \ell_{a,k}(t) &= p\left(X_a(W_t) \mid \tilde{X}_a(W_t; \theta_k), X_j(W_t), \mathcal{M}\right), \\ \pi_{a,k}(t) &= \frac{\ell_{a,k}(t)}{\sum_m \ell_{a,m}(t)}, \quad \hat{\theta}_a(t) = \sum_k \pi_{a,k}(t) \theta_k. \end{aligned} \tag{2}$$

where W_t is the interaction window up to time t , X_j is the counterpart trajectory and $\mathcal{M}$ is the map and reference geometry. The IPV is parameterised as $\theta \in [-\pi/2, \pi/2]$, with positive values placing more weight on the interaction term. Each candidate trajectory for preference θ_k optimises the frozen implementation’s utility $\cos \theta_k \cdot (\text{own-progress cost}) + \sin \theta_k \cdot (\text{interaction cost})$, so a more negative θ weights the joint term negatively — the assertive direction; the full cost terms and solver configuration are released with the code. This is inverse planning, not a single joint maximum-likelihood optimum; $u_a(t)$ is a dispersion statistic of the normalised candidate weights, not a standard deviation, and the operational readability gate $u_a(t) < u_0$ of Eq. (3) is operationalised by the following frozen rule: the reading is discarded as near-uniform when the largest normalised weight falls below 0.20 (against $1/7 \approx 0.14$ for exactly uniform weights) or the candidate likelihoods are exactly tied. The reference frame is what makes the estimate

usable online: against an observed-trajectory-prefix reference the causal current estimate agrees only weakly with the offline estimate (correlation ≈ 0.281), whereas against a static map-lane-centreline reference, route-conditioned and available before the outcome, it agrees closely (correlation ≈ 0.993 , MAE ≈ 0.027), bounded to the lane-referenced slice (about three quarters of cases) and stated as a non-anticipation property, not a causal-effect claim. The gate determines whether the frozen candidate set yields a readable trajectory-based score. The frozen configuration uses seven candidate preferences $\theta_k \in \{-3, -2, -1, 0, 1, 2, 3\} \times \pi/8$; the grid spans $[-3\pi/8, 3\pi/8]$ inside the declared domain $[-\pi/2, \pi/2]$ of Eq. 2, and the admissible span quoted in Fig. 3 is the candidate span $3\pi/4$ (Fig. 3 caption). A rolling window of the ten most recent frames at 10 Hz (one second), with at least four observed frames required for a reading; and a Gaussian trajectory likelihood $\ell_{a,k} \propto \exp\{-\text{MSE}_k/(2\sigma^2)\}$ with $\sigma = 0.1$, where MSE_k is the mean squared Euclidean distance (m^2) between observed and candidate positions over the window, so σ is in metres. The reported reading is not one of the seven candidates: the likelihoods are normalised to weights $w_k = \ell_{a,k} / \sum_j \ell_{a,j}$ and the reading is their weighted mean $\hat{\theta}_a = \sum_k w_k \theta_k$, which varies continuously over $[-3\pi/8, 3\pi/8]$. The candidate set is the support of the likelihood, not the range of the estimator; the concentration of w is what the readability gate below tests. A reading failing the near-uniform rule above carries no discriminative information between the candidates, and the monitor abstains. The full frozen configuration is released with the code.

4.2 Interaction opportunity, readability and episode summary

An interaction-opportunity indicator $o_{ij}(t)$ marks when a behaviourally meaningful interaction is active (a mapped conflict point reachable by both agents within a bounded horizon with closing geometry), as opposed to mere co-presence. Candidate-weight concentration supplies the operational candidate-discrimination statistic $u_a(t)$. The readability indicator is

$$g_a(t) = \mathbf{1}[o_{ij}(t) = 1, u_a(t) < u_0, q_a(t) > q_0, c_a(t) > c_0], \quad (3)$$

where $q_a(t)$ records trajectory/reference information and $c_a(t)$ counterpart-identity stability. The thresholds are frozen on training/guard data and applied identically downstream; $g_a(t) = 0$ states are labelled as no active interaction, insufficient candidate discrimination, unstable identity or unreliable reference rather than scored. The event-level summary of an interaction is the mean current IPV over its valid interaction segment; alternative rules (all valid frames; readability-weighted) differ by 0.26 rad on average and flip the episode sign in 7–22% of episodes (Supplementary Fig. S1c), which is why the monitor never uses an episode summary online and every reading is per-moment.

Three dyad-level analyses in Section 2.2 use the episode summary for both partners. The cohort is the 34,850 two-agent cases with a valid estimate for both partners. Complementarity treats the two summaries as an exchangeable pair and compares their intraclass correlation with a null that re-pairs partners across cases within 415 scenario-matched strata (source subset, path-geometry category, post-encroachment band and duration band); 34,757 cases lie in strata that support re-pairing. The estimate is computed in two independently built implementations and uncertainty is obtained by scene-cluster bootstrap. Early legibility predicts, out of fold under grouped cross-validation, which partner’s summary will finish on the giving-way side from the first fraction of the interaction or from early kinematics over the same window; discrimination is AUC with scene-cluster bootstrap intervals. Turning yield is the paired difference between the turning and straight-driving partners, reported by source and pooled with Benjamini–Hochberg correction across turn groups.

4.3 Context-conditioned human reference range

The reference range is the conditional distribution of the human current IPV given the current observable situation z_t (geometry, role, interaction progress, relative kinematics, online-computable risk proxies, and support/readability state), estimated by conditional quantile regression (histogram-based gradient-boosted quantile models, one per quantile level, with frozen hyperparameters: learning rate 0.06, 72 boosting iterations, maximum 31 leaf nodes, minimum 80 samples per leaf, L2 regularisation 0.01, 255 bins, no early

stopping) and evaluated only on readable, active windows. The reference pool contains human–human vehicle pairs only (2,442,625 anchor rows): any interaction in which either agent is an automated data-collection vehicle is excluded from the reference, so machine behaviour cannot contaminate the human range. The corpora are fleet-collected in specific cities in the United States and Singapore; the reference is a sample of natural driving there, not of all human driving. We compare a global reference, an offline oracle-risk ceiling, and the context-conditioned reference; the context-conditioned reference supplies essentially all of the achievable sharpening over the global reference (width reductions of 21%, 20% and 8% at the 80%, 90% and 95% levels, at coverage within 0.6 percentage points of nominal; the global 95% range spans essentially the whole admissible scale). Two further inputs are evaluated as ablations and add no measurable value: conditioning on the principal counterpart’s current IPV, and conditioning on the ego agent’s own behavioural history (the self-anchor). Neither is interpreted as a mechanism; the counterpart channel in particular is statistically indistinguishable from removing the IPV input given the situation (paired 90% interval-score difference -0.0002 , case-clustered $p = 0.86$), which—set against the event-level partner correlation—is the tension highlighted in the Results. Concretely, z_t comprises four categorical descriptors (path-geometry category, path relation, turn-pair label and priority role) and 22 numeric channels: elapsed interaction time and history length; ego and counterpart velocity components and headings at the anchor; relative position components, distance, velocity components, speed, closing rate and heading difference at the anchor; short-window means and a dispersion of relative distance, relative speed and closing rate; and two online risk proxies — the closing-rate time-to-collision at the anchor and an anticipated post-encroachment proxy computed only from information available at time t . Excluded from the reference are observed post-encroachment time, realised passing order, post-hoc phase labels, the episode summary, the machine/human identity, and target-proximal concurrent ego actions; quantiles are made non-crossing by rearrangement. The reference range conditions on the current situation rather than on a specific temporal pattern of the interaction; temporal-history features were examined and are not part of the primary conditioning.

4.4 Conformal calibration and abstention

Vocabulary, used consistently: an anchor row is one agent-moment eligible for estimation; a readable moment passes the discriminative-information rule; an accepted moment additionally lies within human support (these receive verdicts); on the benchmark we write candidate moment for an anchor row entering the two gates, and judgeable moment for one that passes both. A scene is one contiguous recorded segment and the unit of the frozen split; an interaction case is one interacting vehicle pair within a scene (Section 2.2); on the benchmark, a scenario is one of the staged templates and a scenario run is one system’s or one driver’s traversal of it (Section 2.4). The final interval is calibrated by split-conformal prediction [49, 51] with nonconformity $s_k = \max\{\hat{q}_l(x_k) - y_k, y_k - \hat{q}_u(x_k), 0\}$ and radius $c_\alpha = s_{(\lceil (n+1)(1-\alpha) \rceil)}$ over accepted calibration moments. Conformal calibration of trajectory atypicality has precedent in unconditional surveillance settings [52]; distribution-shift-aware conformal variants [53–55] are compatible with the same monitor, and we use the split-conformal form for auditability. The estimand is pointwise marginal coverage over accepted moments: the conformal radius c_α is the $\lceil (n+1)(1-\alpha) \rceil$ -th smallest nonconformity score over all accepted calibration-fold moments ($n = 455,723$), and coverage is evaluated over all accepted test-fold moments (461,937). The fitted radii are near zero ($c_\alpha = 1.4 \times 10^{-3}$, 1.2×10^{-6} and 0.0 rad at the 80%, 90% and 95% levels), so the conformal step finds essentially nothing to repair in the situational quantile model. Splits are made by whole scenes under a split frozen before this study, so no scene contributes to more than one fold; the readability gate is frozen on training/guard data and applied identically to calibration and test. Moments within an interaction are dependent, so the coverage statement is marginal over accepted moments — not a per-interaction, sequential or conditional guarantee — and the achieved coverage is reported empirically (Fig. 3b). A marginal statement leaves conditional coverage unconstrained in principle, so we also report it by source: on the test fold the deployed reference contains 88.2%, 95.9%, 85.9% and 88.2% of moments at the 90% level within Waymo, nuPlan, Lyft and Argoverse-2 respectively, against 90.3% pooled, so the pooled figure is not masking a source in which the range fails. The frozen split assigns 974,984 / 499,893

/ 481,088 / 486,660 human anchor rows to the train / guard / calibration / test folds (26,828 interaction cases; whole-scene split frozen in advance). The universes nest as follows: of the 38,228 corpus interaction cases (Section 2.2), 26,828 enter the frozen anchor ledger, contributing 4,497,368 anchor rows (the readability accounting above); the human-human reference pool retains 2,442,625 of those rows (Methods 4.3), which are exactly the train/guard/calibration/test folds ($974,984 + 499,893 + 481,088 + 486,660$). The support gate scores each moment by the mean Euclidean distance to its $k = 25$ nearest training anchors after standardisation and one-hot encoding of the situation categories, abstaining above the 95th percentile of guard distances (1.081); categorical support additionally requires at least 50 training anchors and at least 10 cases. On the test fold, support abstention is 5.08% (24,723 of 486,660) with zero categorical failures. On the natural corpus the mechanism-level readability accounting over 4,497,368 anchor rows is: 3,202,646 readable, 1,275,480 near-uniform weights, 17,416 exact ties and 1,826 solver failures. The 1,826 solver failures belong to this natural-corpus accounting; on the benchmark, no candidate moment failed the solver or pipeline (Section 2.4). Conditional coverage across state groups and trajectory-wise simultaneous coverage are not established here and remain stated boundaries of the construction; leave-one-source-out coverage quantifies the source-shift boundary: at the 90% level, refitting with one source held out covers the held-out source at 0.743 (Waymo, 143,380/193,096), 0.990 (nuPlan, 149,068/150,587), 0.750 (Lyft, 68,270/91,069) and 0.900 (Argoverse-2, 11,315/12,576, with 44.3% held-out abstention—the support gate intercepting unsupported situations); transfer across sources is therefore a bounded property of source heterogeneity, and coverage and abstention must be read together. We report not only accepted-window coverage but also the active-interaction rate, readability rates, and the total abstention rate with its reason distribution, so that filtering is never hidden. The monitor returns ABSTAIN—not WITHIN-NORM—with one of the reason codes NO-ACTIVE-INTERACTION, EGO-UNREADABLE, COUNTERPART-UNREADABLE (the counterpart’s identity is not stable enough to anchor a reading, c_a of Eq. 3), MAP-OR-ROLE-UNCERTAIN, LOW-HUMAN-SUPPORT or OUT-OF-DISTRIBUTION; reason codes are assigned by the first failing gate

in the evaluation order of Algorithm 2, so exactly one code is returned; otherwise it emits WITHIN-NORM, COMPETITIVE-DEVIATION or OVER-YIELDING. In the main text, COMPETITIVE-DEVIATION is written ‘outside the range on the assertive side’ and OVER-YIELDING ‘outside on the accommodating side’; both sides are tested at every judgeable moment. A sequential warning layer (persistence over consecutive verdicts) belongs to the deployment interface: no result in this paper uses it, every reported verdict and flag rate is per-moment, and its operating characteristics (episode-level false-alarm rate, detection delay, warning duration) are not evaluated in this paper.

4.5 Real-vehicle consequence protocol

We use a matched-scenario real-vehicle benchmark at the scenario level (no run-level, seed-level or causal claims), separating an outcome universe from a replay/IPV universe with frozen exclusions. The counterpart vehicles are driven by traffic-simulation software (TESS NG) and respond to what the ego vehicle does; the same counterpart control is used in the human reference arm below, so the two arms face a counterpart that reacts to them on the same terms. The staging is mixed-reality: the ego is a real vehicle driven at a test site, and the simulated counterparts reach an automated system through its perception interface and a human driver through a head-mounted display. No confirmatory analysis proceeded until an outcome-blind crosswalk had reconciled canonical scenario identifiers, structural case identities and official outcome labels. Realised-consequence events are defined by a frozen automatic extractor. Verdicts are issued per moment; outcome windows begin at the flagged moment, so the trajectory samples that produce a verdict (the one-second window ending at that moment) precede every outcome sample. The ego-side outcome is the minimum time-to-collision to the counterpart over the fixed three-second post-verdict window (30 frame intervals at 10 Hz, inclusive of both endpoints; the run ends inside that window for 4.8% of assertive, 6.2% of accommodating and 5.0% of within-range moments, and the window is then truncated at the last available frame); time-to-collision is computed per frame as distance over closing rate, and frames in which the pair is not closing do not enter the minimum; the counterpart-side

outcomes are read over the same fixed three-second window from the same moment, from the other vehicle’s own logged control record. Comparisons are made within the frozen situation cells, inference resamples scenario runs (the unit of clustering throughout), and a placebo test that reassigns whole exposure trajectories across scenario runs distinguishes the real flag timing from permuted timings (200 whole-trajectory reassignment draws; the comparison statistic is the absolute case-clustered t ; empirical $p = 0.0199$). Across all six level–window combinations of the $\log(1 + \text{TTC})$ magnitude battery the coefficients agree in sign; the fixed three-second window is the pre-specified primary and is the one displayed, and its case-clustered intervals exclude zero at all three levels, while the alternative and much shorter window—the anchor’s own prediction-target horizon, a median 0.60 s—crosses zero at the 90% level ($[-2.6100, +0.1372]$). A case-level label permutation on the same battery does not reach significance ($p = 0.1493$); the exposure placebo above is the test specific to flag timing. The battery is reported for both sides of the range. Its endpoints, windows, thresholds, clustering unit and resampling settings were fixed for the assertive-side comparison and were not altered afterwards; the accommodating-side values were obtained by running that unchanged battery on the accommodating group, so no endpoint, threshold or window was chosen in the light of accommodating-side data. The counterpart braking endpoint was specified for both sides from the outset. On the accommodating side (817 ego moments and 719 counterpart anchors, against the same within-range groups) the counterpart’s speed reduction is $0.79\times$ the within-range median ($[0.28, 1.53]$) and its speed variation $1.08\times$ ($[0.78, 1.53]$); the share of counterpart records braking below -3 m s^{-2} differs by $+0.07$ percentage points ($[-1.96, +2.24]$); and the ego’s own margin median is $0.99\times$ ($[0.83, 1.22]$) with its upper quartile at $1.03\times$ ($[0.67, 1.38]$). Every interval admits no difference. These are non-detections at the achieved precision—which on the ratio scale is slightly coarser than on the assertive side, despite the larger group—and not demonstrations of equivalence. The two sides are additionally compared with each other, rather than each against the within-range group in turn: on every re-sampled set of scenario runs both ratios are recomputed and their difference taken, so the two sides move together and the comparison keeps the correlation that separate inter-

vals discard. At the upper quartile the sides differ (-0.43 $[-0.71, -0.06]$, $p = 0.014$); at the median they do not (-0.09 $[-0.29, +0.12]$, $p = 0.368$). That ordering is unchanged on the shorter window and under the retention convention above—the upper-quartile difference excludes zero in all four window-by-convention combinations and the median difference in none—so the side-specific statement is made at the quartile that supports it and withheld at the one that does not. Estimates are weighted by moment rather than by scenario run, because the monitor issues a verdict at each moment and the per-moment population is the one its operation defines; resampling scenario runs sets the intervals. Weighted equally per run, the assertive-side counterpart braking contrast does not reach significance ($p = 0.4704$); the speed-reduction and speed-variation contrasts carry the counterpart-side result. Ratios of medians are reported alongside differences because a ratio becomes unstable where the within-range median is near zero, as it is in one of the two sites. Of the 14,099 judgeable moments at the 90% level, 519 lie below the range, 869 above and 12,711 inside. The criterion is the situation-conditioned range, not the sign of the reading: at the 90% level the lower edge is negative in essentially every situation (median -1.03 rad), 40.0% of within-range moments also carry a negative reading, and of all negative-reading moments only 9.25% are flagged. The challenge fielded 20 systems; one is set aside in full because its replay records are not clean, leaving 19 systems across the 15 scenarios, or 285 system–scenario cells, of which 267 yield the trajectory anchors the estimator needs. The 18 that do not were all attempted and officially scored (15 scored completions and 3 collision failures); they are lost in our own replay processing — 11 to observed reference trajectories that collapse to fewer than two distinct points, 3 to no timing-valid anchor, 3 to no eligible counterpart and 1 to a counterpart without timing support — and 13 of the 18 fall in a single system. No scenario is dropped: each retains between 16 and 19 systems. Consequence analyses follow the same run accounting as the rest of the benchmark. Of the 267 scenario runs, 231 contain at least one judgeable moment and can therefore enter a consequence analysis at all. Reported per run rather than per moment, 157 of those 231 (68.0%) contain at least one flagged moment on either side of the range and 120 (51.9%) at least one on the assertive side.

The per-moment rate is the operating point throughout; the per-run figure is given here because the two have different denominators and are easily confused. Of those, 27 are runs in which the counterpart was identified from a different naming convention in the run logs; they are held out of the pooled counterpart-response analyses and reported separately, leaving 204, of which 175 have the recorded counterpart logs the outcome battery requires. The split is one of record-keeping rather than of apparatus: the counterpart is the same kind of simulator-driven vehicle throughout. The runs dropped at that last step do differ from the retained ones in how often they contain a flagged moment: 7 of the 29 dropped (24.1%) against 132 of the 175 retained (75.4%). The requirement is a logging one, and a run without recorded counterpart logs also tends to be one with fewer judgeable moments, so the two rates are not independent of each other; we report them rather than argue the point. The restriction applies only to the counterpart-side panels: the ego-side panels use all 227 runs with an evaluated ego window and are not filtered on counterpart logging at all. That the counterparts respond is also visible in the replay itself: in the four scenarios whose counterpart identity recurs across systems, the same counterpart’s trajectory differs between systems by 4.35–9.03m, and in none of the four is it identical. The ego-margin panels use the moments with a defined post-verdict margin (499 assertive, 817 accommodating, 12,344 within-range). The margin is undefined where the pair is not closing, and the three groups do not lose equally: 3.9% of assertive moments (20 of 519), 6.0% of accommodating moments (52 of 869) and 2.9% of within-range moments (367 of 12,711). The accommodating group loses most, which follows from what defines it—yielding is what stops a pair closing—so the exclusion removes preferentially the moments in which that group is most itself, and it does so in the direction that draws the accommodating group toward the within-range one. We therefore report the battery again with those moments retained and ranked at the safe end rather than dropped, which is what never approaching a collision means; the quartile statistics are invariant to the value used, verified against two sentinels a thousandfold apart. Retaining them leaves the assertive upper-quartile contrast in place (0.626 of the within-range quartile, [0.410, 0.829], against 0.601 [0.436, 0.831] when they are dropped)

and moves the accommodating side further away from the within-range group in the direction of wider margins (1.029 to 1.126), which is what restoring the most strongly yielding moments should do. The exclusion was therefore pulling the accommodating side toward the within-range group, not the assertive side. Under these definitions the harm relationship is bounded rather than confirmed: deviation does not robustly or specifically predict realised harm, and the dominant limitation is upstream, in the adequacy of interaction-failure segmentation itself. The fraction of moments whose reading carries discriminative information differs between the naturalistic corpora and the benchmark (70.3% vs 55.3%); we report this as an observed difference without attributing a cause, and the readability gate treats both populations identically. The binding constraint on benchmark coverage is situational rather than volumetric: in the 36 always-silent runs the readability pass rate (53.78%) is close to the benchmark average (55.30%) while the human-support pass rate is 0.07% (against 32.32% overall), and support does not follow sample volume — one high-volume situation cell with 1,148,133 human anchor rows passes support on 14.58% of its benchmark moments while a 45,283-row cell passes on 47.03%. The support gate compares kinematic neighbourhoods, not IPV values.

4.6 Human reference arm (matched scenarios)

Licensed drivers drove the same scenario set in a real vehicle against the same simulator-driven counterparts under the counterpart control described above, giving a drivers-by-scenarios matrix (20 drivers $\times$ 15 scenarios) symmetric to the automated-systems benchmark. The unchanged estimator, abstention gates and frozen reference range were applied to the human runs. Endpoints were fixed before analysis: (i) the flagged fraction of judgeable moments against the nominal level, with the natural-driving outside rate (9.72% at the 90% level) as the apparatus-validity anchor; and (ii) the consequence battery of Fig. 4 recomputed within the human arm under identical frozen definitions, with no new outcome variables and no threshold scans. Official competition scores, harm labels and preference ratings were excluded as endpoints in the analysis plan, which was specified and frozen before any outcome was examined; those labels are publicly released by the

challenge, which is operated by the authors’ research group, and no author entered the challenge as a competitor. Driver identities were pseudonymised before analysis.

The human-reference study was approved by the Science and Technology Ethics Committee of Tongji University (approval tjdxsr2025011). All participants gave written informed consent before taking part and were insured for the duration of testing. Runs took place at a dedicated test site with a safety supervisor on board who could intervene or terminate a run at any time. Participants held a valid driving licence, were compensated at CNY 150 per hour, and were instructed to drive naturally; they were not told that their runs would serve as a human reference against automated systems. Each driver completed the scenario set as a single fixed sequence, identical across drivers, and no trial was missing.

4.7 Subjective evaluation of selected interaction segments

Cohort and stimuli. The author-confirmed final rating dataset contains 40 participants organised into 20 pairs and 40 sessions. Of 600 formal trial attempts, 596 were completed and four were aborted. Each completed trial contributes one ego-position and one counterpart-position rating record, yielding 1,192 records; repeated records from the same participant are not treated as independent participants. The 90 segments span 15 scenarios: 24 assertive-side segments, 36 accommodating-side segments and 30 within-range comparison segments, contributing 158, 239 and 199 completed trials, respectively. The supplied source labels identify 45 automated and 45 human segments. These labels are used for sensitivity summaries, not a source-recognition or vehicle-ranking claim. The subjective protocol is distinct from the 20-driver matched driving arm.

The supplied segment categories were retained. Package-internal checks associate A with segments containing below-range moments, C with segments containing above-range moments, and W with within-range and abstention moments. A segment category does not assign the same verdict to every frame; abstention is never recoded as a within-range verdict. Stimulus-selection rules and the version relationship between these supplied labels and the manuscript’s frozen monitor require completion in the study documentation

below.

Rating items. Analyses are separated by the supplied ego and counterpart position labels. Perceived atypicality is Q1 reverse-coded as $8 - Q1$; Q4 is comfort, with higher values indicating greater comfort; Q5 is the supplied reported interaction-cost item, with higher values indicating greater cost. These items use a 1–7 scale. Q5 and Q5b are structurally unavailable for the ego position: the 596 blank values for each item there are not zero ratings. The four aborted trials contribute no completed questionnaires and are not imputed. Supplementary Note 2 records the remaining exploratory items and the scope of their interpretation.

Statistical analysis. This is an exploratory secondary analysis of the final rating cohort. Within each participant and position, ratings were averaged within category; category differences were then computed within participant and averaged with equal participant weights. Pointwise 95% percentile intervals use 10,000 bootstrap resamples of the 20 participant pairs, keeping the two members together. Two-sided one-sample t -tests use the 20 pair-averaged differences (19 degrees of freedom). Holm correction covers all 30 available item-by-position-by-category contrasts, including contrasts not highlighted in the main text. Bootstrap intervals are pointwise, not multiplicity-adjusted simultaneous intervals, and condition on the selected stimulus set. The extraction used Python 3.9.6, NumPy 2.0.2, pandas 2.3.3 and SciPy 1.13.1, with seed 20260925.

A separate scenario-composition sensitivity analysis first averages ratings within segment, then segments within scenario and category, and finally contrasts the categories across shared scenarios with equal scenario weights. Assertive-minus-within, accommodating-minus-within and assertive-minus-accommodating comparisons contain eight, nine and 12 shared scenarios, respectively. Its intervals resample scenarios and condition on the observed raters; they are not a joint participant-and-stimulus uncertainty estimate. The participant-pair and scenario analyses address different sources of variation and are not independent replications.

[PLANNED: Complete the subjective study’s recruitment and eligibility, apparatus and presentation instructions, allocation and order rules, compensation, ethics approval and

Algorithm 1 Offline calibration of the context-conditioned reference range

Require: existing IPV time series $\mathcal{D}$; target coverage $1 - \alpha$ **Ensure:** quantile model $\hat{Q}$, conformal radius c_α , frozen readability gate

- 1: split whole scenes into train / guard / calibration / test
 - 2: freeze the interaction-opportunity rule and the readability gate (u_0, q_0, c_0) on train/guard
 - 3: **for** each interaction in train **do**
 - 4: keep active, readable windows; build the current situation z_t ; target $y \leftarrow$ human current IPV
 - 5: **end for**
 - 6: fit non-crossing situational quantiles; fit counterpart-IPV and self-history variants as ablations only
 - 7: apply the identical frozen gate to calibration; score every accepted calibration window
 - 8: $c_\alpha \leftarrow s_{(\lceil (n+1)(1-\alpha) \rceil)}$ over accepted calibration windows; freeze all parameters
 - 9: **return** $\hat{Q}$, c_α , gate {evaluate once on test}
-

Algorithm 2 Online social-atypicality monitoring step at time t

Require: window W_t ; map-lane reference; calibrated $\hat{Q}, c_\alpha$; frozen gate; source health**Ensure:** reference range and verdict, or an abstention with reason code

- 1: identify the principal counterpart; evaluate active interaction $o_{ij}(t)$
 - 2: **if** $o_{ij}(t) = 0$ **then**
 - 3: **return** ABSTAIN(No-Active-Interaction)
 - 4: **end if**
 - 5: estimate $\hat{\theta}_i(t)$ and reliability $u_i(t)$ {Eq. (2)}
 - 6: **if** $g_i(t) = 0$ **then**
 - 7: **return** ABSTAIN(EGO-UNREADABLE if u_i or q_i failed; COUNTERPART-UNREADABLE if c_i failed)
 - 8: **end if**
 - 9: build the current situation z_t ; run map/role, human-support and distribution gates
 - 10: **if** any gate fails **then**
 - 11: **return** ABSTAIN(reason code)
 - 12: **end if**
 - 13: $[\hat{q}_l, \hat{q}_u] \leftarrow \hat{Q}(z_t)$ widened by c_α
 - 14: verdict $\leftarrow$ WITHIN-NORM / COMPETITIVE-DEVIATION / OVER-YIELDING from $\hat{\theta}_i(t)$ vs $[\hat{q}_l, \hat{q}_u]$
 - 15: optionally map the verdict to a planner-facing signal through a persistence rule (deployment interface; not evaluated in this paper)
 - 16: **return** (range, verdict) {plus signal only where the optional persistence rule is instantiated}
-

consent scope, full questionnaire wording and anchors (especially Q5), stimulus-selection rules, and the version link between supplied labels and the frozen monitor. These records were absent from the extraction package; do not substitute the matched human-driving protocol.]

Planner interface. The verdicts map to planner actions—audit, soft cost or warning, a fallback candidate under sustained competitive deviation at high physical risk, a hesitation/efficiency warning, and — on abstention — a degraded mode in which the social verdict is simply unavailable. This mapping is an interface specification, not an external-validity result; the social layer is advisory and subordinate to collision-safety enforcement, no social verdict or abstention relaxes or delays any safety constraint, and a planner fallback is a control action never labelled WITHIN-NORM.

Data Availability

InterHub-derived interaction cases and source datasets (Waymo, nuPlan, Lyft, Argoverse-2) are governed by their respective licences; processed derivatives, including the online IPV time series, will be made available upon publication. The matched-scenario real-vehicle benchmark is operated by the authors’ research group at Tongji University, and its official outcome labels are publicly released; the derived moment-level verdict series for the automated-systems arm will be made available upon publication. Records from the human reference arm are not released: participants consented to take part in the study, and that consent does not extend to sharing or publishing their driving records, so the human arm is reported only in aggregate.

[PLANNED: Specify the separate subjective study’s data-access and sharing terms from its own consent records, including what may be released as aggregate Source Data and whether any participant-level derivatives may be shared. The driving-arm restriction above does not establish the subjective study’s permissions.]

Code Availability

Implementation, the online likelihood-based inverse-planning IPV estimator, the readability gate and abstention logic, and figure-generation code will be deposited in Zenodo under a persistent identifier upon publication, and are available to referees on request during peer review.

Supplementary Information

Supplementary Note 1. Published validation and present implementation of IPV estimation

The IPV estimator used in this study was introduced in Zhao et al. [24]. In controlled VGIM interactions, agents were assigned interaction-preference patterns and the trajectory-based estimate tracked those imposed patterns as the interaction unfolded; incorporating the identified IPV also improved interactive-motion prediction. The present study adopts that published estimator without refitting it and changes the downstream question from parameter identification to membership in a situation-conditioned human reference range.

The online implementation evaluates seven candidate preferences $k\pi/8$, $k \in \{-3, -2, -1, 0, 1, 2, 3\}$ over a one-second causal window at 10Hz. Each candidate generates a trajectory under the frozen VGIM utility, the observed-to-candidate trajectory error is converted to a normalised likelihood, and Eq. 2 returns their continuous weighted mean. A static route-conditioned map reference keeps the calculation causal and available at decision time. The operational readability gate asks whether the current candidate set is sufficiently discriminated while the interaction, trajectory reference and counterpart identity are valid.

The present implementation was also checked against interaction-specific controls (Supplementary Fig. S1). Candidate discrimination sharpens in the actual dyad relative to time-misaligned, partner-swapped and no-opportunity controls. At the same time, readable frame-level values can continue to move as negotiation develops, and episode-level summaries depend on how valid frames are aggregated. These diagnostics support the per-moment design used throughout the paper: the monitor retains the frame-level reading, applies an explicit readability boundary and never substitutes a single agent-level score for the online trajectory.

Table S1. Role of IPV estimation in the published T-ITS study and in the present monitor.

Aspect	Published T-ITS study [24]	Present study
Scientific role	Introduces trajectory-based IPV identification in the VGIM framework	Uses the published estimator as a fixed input to a human-reference monitor
Interaction setting	Controlled pairwise VGIM interactions	Naturalistic and matched-scenario pairwise interactions
Output	Time-varying IPV identified from observed interaction trajectories	Frame-level IPV reading compared with the human range for the current situation
Primary question	Can imposed interaction-preference patterns be identified and improve interaction prediction?	Does the current behaviour fall within the context-conditioned human reference range?
Validation emphasis	Model-consistent identification and interactive-motion prediction	Context conditioning, calibration, abstention and behavioural auditing

Supplementary Note 2. Subjective endpoints and exploratory comparisons

The main subjective results focus on perceived atypicality, comfort and reported counterpart interaction cost. All 30 item-by-position-by-category comparisons were retained in the extraction, including the supplied Q2 yielding-tendency item, Q3 predictability item and counterpart-only Q5b naturalness item. Their full wording and response anchors must accompany the final questionnaire. No item is substituted for a direct preference, acceptability or appropriateness question.

For ego-position atypicality, the assertive-minus-within difference is $+0.465$ $[+0.170, +0.759]$ points, with Holm-adjusted $p = 0.0704$. It is therefore not described as a multiplicity-adjusted positive finding despite its pointwise interval excluding zero. The counterpart-position assertive-minus-accommodating atypicality comparison has adjusted $p = 1$; this non-detection does not establish equal atypicality. Q2 does not clearly separate assertive from within-range segments, and the Q5b comparisons do not establish equivalence of naturalness. Continuous dial responses are not used as evidence of instantaneous discomfort or warning latency: their construct, anchors, standardisation and onset definitions require additional documentation.

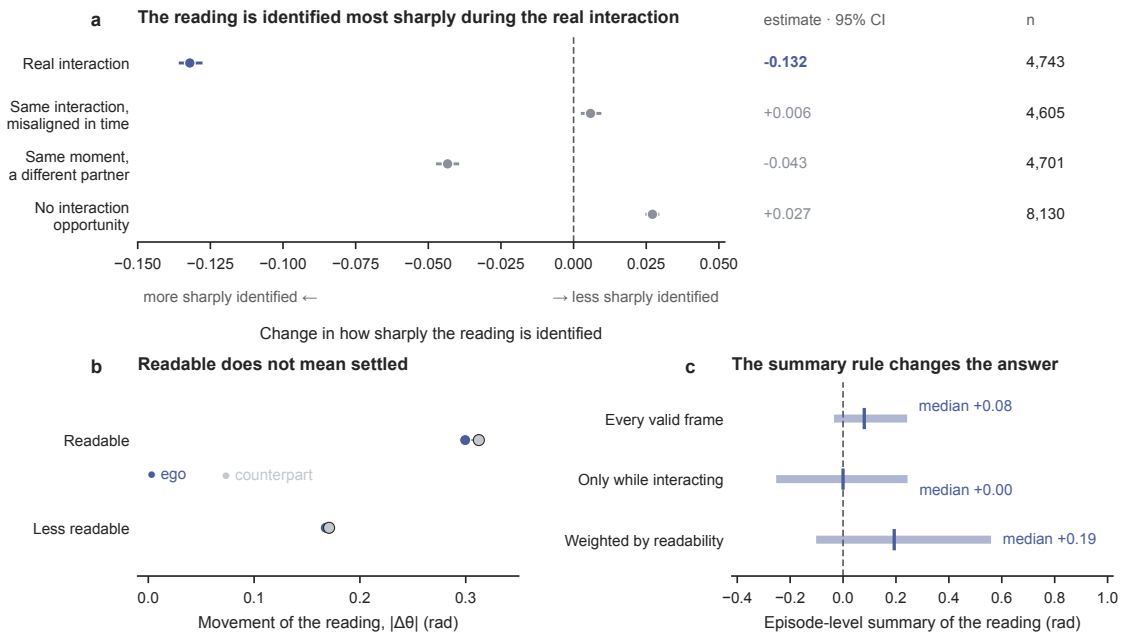


Figure S1. Interaction-specific diagnostics and summary sensitivity of the present online implementation. **a** Change in candidate-weight concentration inside real interaction windows relative to three controls. The interaction-specific sharpening disappears under time misalignment (+0.006), is weaker under partner swapping (−0.043), and reverses where no interaction opportunity exists (+0.027). Points are means with 95% confidence intervals over case clusters; the n column counts case clusters. **b** Mean movement of the frame-level reading between consecutive frames. Readable values remain dynamic as an interaction unfolds, so the monitor keeps them time-indexed. **c** Episode-level summaries of the same interactions under three aggregation rules; bars span the interquartile range and ticks mark medians. The rules differ by 0.26 rad on average and change the sign in 7–22% of episodes, motivating the per-moment verdict used in the main analysis. Source data are provided as a Source Data file.

Table S2 gives the eight counterpart-position contrasts highlighted in the Results. Participant-pair intervals condition on the selected stimuli. The common-scenario estimate uses equal weighting of segments within scenario and then of shared scenarios; it is a composition sensitivity summary with a different estimand, not a second experiment.

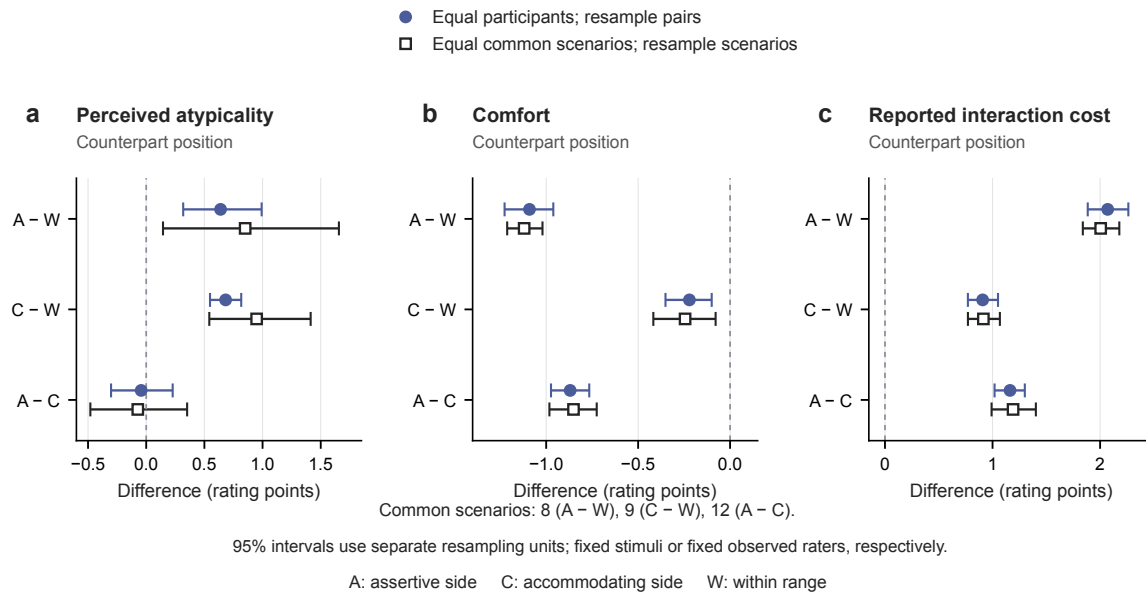


Figure S2. Subjective contrasts under participant-paired and common-scenario weighting. Counterpart-position contrasts for **a** perceived atypicality, **b** comfort and **c** reported interaction cost. Filled circles reproduce equal-participant within-participant differences with participant-pair bootstrap intervals; open squares give equal-common-scenario estimates with scenario bootstrap intervals. These are distinct estimands with distinct resampling units. A-W, C-W and A-C use 8, 9 and 12 shared scenarios, respectively. All intervals are the existing pointwise 95% intervals from 10,000 draws. Scenario resampling fixes the observed raters; participant-pair resampling conditions on the selected stimuli. This is a composition sensitivity analysis, not an independent replication. Source Data retain all 30 contrasts.

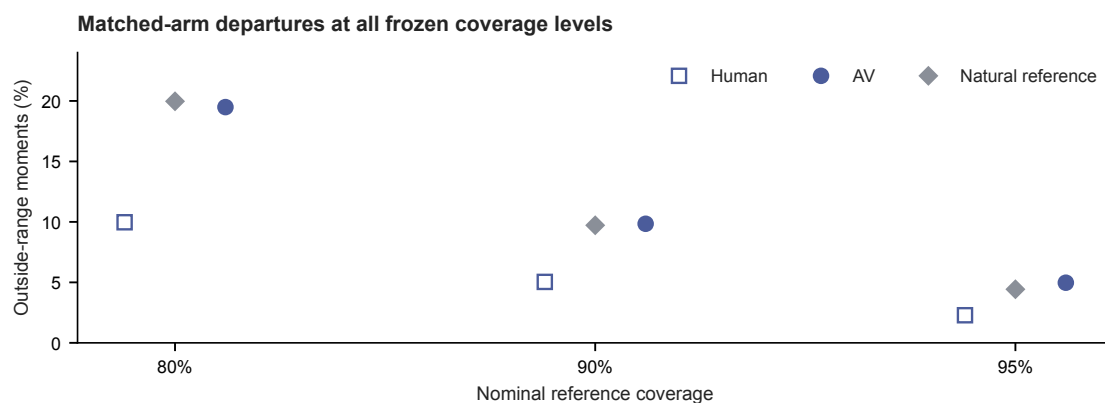


Figure S3. Matched human and automated flag rates across the three frozen reference levels. Outside-range rates at nominal 80%, 90% and 95% coverage for the natural-driving test reference and the two matched benchmark arms. Denominators are judgeable moments in the corresponding population. The natural reference is an alarm-inflation check and is not a matched comparison of behavioural levels. The 90% operating point remains the prespecified main display; the other two levels are retained here without selection by significance. Points are frozen estimates; no missing interval is reconstructed. Source Data provide exact counts, rates and available intervals.

Table S2. Principal counterpart-position subjective contrasts. Differences are on 1–7 scales; positive atypicality and cost differences indicate higher ratings, while negative comfort differences indicate lower comfort. A, assertive-side; C, accommodating-side; W, within-range comparison. All rows use 40 participants in 20 pairs. Intervals are pointwise 95% participant-pair bootstrap intervals. Adjusted p values use the full 30-comparison Holm family. Common-scenario means use eight A–W, nine C–W and 12 A–C scenarios.

Item	Contrast	Difference [95% CI]	Holm p	Scenario mean
Atypicality	A–W	+0.639 [+0.318, +0.992]	0.0245	+0.850
Atypicality	C–W	+0.6825 [+0.548, +0.817]	1.99×10^{-7}	+0.949
Comfort	A–W	−1.091 [−1.226, −0.961]	6.11×10^{-11}	−1.120
Comfort	A–C	−0.870 [−0.974, −0.766]	5.18×10^{-11}	−0.851
Comfort	C–W	−0.221 [−0.351, −0.100]	0.0342	−0.245
Reported cost	A–W	+2.070 [+1.885, +2.263]	3.95×10^{-13}	+2.005
Reported cost	A–C	+1.163 [+1.019, +1.300]	6.93×10^{-11}	+1.191
Reported cost	C–W	+0.908 [+0.771, +1.051]	3.49×10^{-9}	+0.915

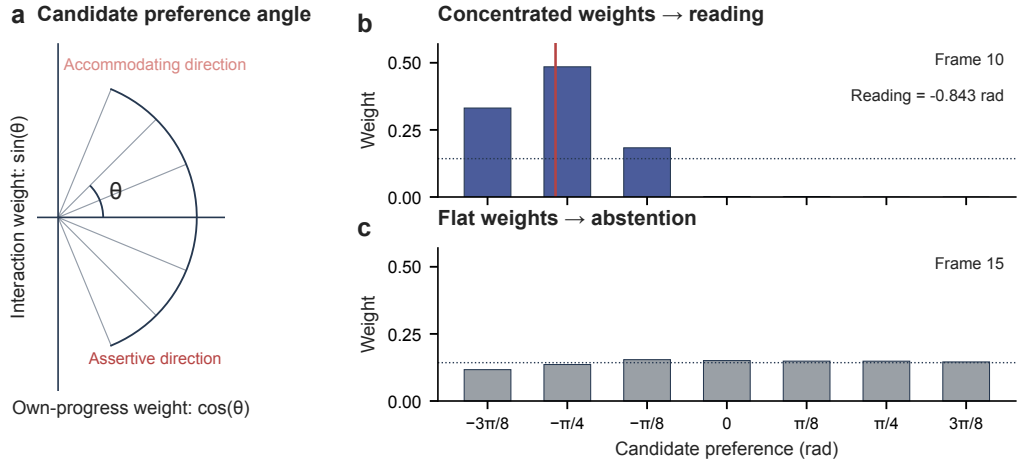


Figure S4. Candidate preferences and the operational readability check. **a** Seven candidate angles weight the agent's own-progress and interaction costs through cosine and sine. They span $[-3\pi/8, 3\pi/8]$ at $\pi/8$ spacing. **b,c** Frozen candidate-weight examples from two frames of one interaction illustrate a concentrated and a near-uniform likelihood distribution, respectively. The readable value is the candidate-weighted mean; the unreadable frame has no reported reading. These panels describe the implementation of the previously published estimator, not a new numerical-recovery guarantee. Source Data contain the original candidate grid and weight vectors.

Supplementary Note 3. Complete aggregate subjective results

These tables reproduce the supplied frozen aggregate outputs without rerunning tests or resampling. A, assertive-side; C, accommodating-side; W, within-range comparison. Ego and counterpart (CP) denote rating positions. Each main contrast uses 40 participants in 20 pairs. Q2 uses its supplied bipolar coding; the remaining items use the supplied rating scales. The main pointwise intervals resample participant pairs 10,000 times. The two-sided tests use 20 pair-averaged differences with 19 degrees of freedom; Holm correction covers all 30 main comparisons. No sensitivity analysis changes the primary inclusion rule or supplies a new corrected significance claim.

Table S3. All 30 paired subjective comparisons.

Position	Item	Contrast	Difference [95% CI]	Raw p	Holm p
Ego	Atypicality	A-W	+0.465 [+0.170, +0.759]	0.007	0.0704
Ego	Atypicality	C-W	+0.756 [+0.612, +0.896]	3.50×10^{-9}	8.76×10^{-8}
Ego	Atypicality	A-C	-0.291 [-0.551, -0.041]	0.0428	0.3422
Ego	Q2 yielding	A-W	+0.013 [-0.339, +0.368]	0.9452	1
Ego	Q2 yielding	C-W	+0.575 [+0.343, +0.800]	1.18×10^{-4}	0.0019
Ego	Q2 yielding	A-C	-0.562 [-0.793, -0.342]	1.30×10^{-4}	0.0019
Ego	Q3 predictability	A-W	-0.679 [-0.854, -0.515]	3.87×10^{-7}	8.12×10^{-6}
Ego	Q3 predictability	C-W	-0.502 [-0.670, -0.324]	2.24×10^{-5}	4.04×10^{-4}
Ego	Q3 predictability	A-C	-0.177 [-0.335, -0.023]	0.0443	0.3422
Ego	Comfort	A-W	-0.666 [-0.802, -0.531]	1.53×10^{-8}	3.37×10^{-7}
Ego	Comfort	C-W	-0.359 [-0.464, -0.255]	2.72×10^{-6}	5.43×10^{-5}
Ego	Comfort	A-C	-0.307 [-0.429, -0.183]	1.28×10^{-4}	0.0019
CP	Atypicality	A-W	+0.639 [+0.318, +0.992]	0.002	0.0245
CP	Atypicality	C-W	+0.6825 [+0.548, +0.817]	8.66×10^{-9}	1.99×10^{-7}
CP	Atypicality	A-C	-0.043 [-0.302, +0.228]	0.7595	1
CP	Q2 yielding	A-W	-0.228 [-0.493, +0.056]	0.1307	0.6535
CP	Q2 yielding	C-W	+0.451 [+0.273, +0.642]	1.89×10^{-4}	0.0025
CP	Q2 yielding	A-C	-0.679 [-0.925, -0.435]	3.69×10^{-5}	6.27×10^{-4}
CP	Q3 predictability	A-W	-0.822 [-0.980, -0.672]	5.30×10^{-9}	1.27×10^{-7}
CP	Q3 predictability	C-W	-0.520 [-0.688, -0.354]	8.46×10^{-6}	1.61×10^{-4}
CP	Q3 predictability	A-C	-0.302 [-0.493, -0.106]	0.0078	0.0706
CP	Comfort	A-W	-1.091 [-1.226, -0.961]	2.18×10^{-12}	6.11×10^{-11}
CP	Comfort	C-W	-0.221 [-0.351, -0.100]	0.0031	0.0342
CP	Comfort	A-C	-0.870 [-0.974, -0.766]	1.79×10^{-12}	5.18×10^{-11}

Position	Item	Contrast	Difference [95% CI]	Raw p	Holm p
CP	Reported cost	A-W	+2.070 [+1.885, +2.263]	1.32×10^{-14}	3.95×10^{-13}
CP	Reported cost	C-W	+0.908 [+0.771, +1.051]	1.34×10^{-10}	3.49×10^{-9}
CP	Reported cost	A-C	+1.163 [+1.019, +1.300]	2.57×10^{-12}	6.93×10^{-11}
CP	Q5b naturalness	A-W	+0.033 [-0.092, +0.146]	0.6128	1
CP	Q5b naturalness	C-W	+0.091 [-0.012, +0.193]	0.1048	0.629
CP	Q5b naturalness	A-C	-0.059 [-0.175, +0.063]	0.3564	1

The source labels below describe the supplied stimulus groups; they do not establish source identification or a vehicle ranking. Intervals use participant-pair resampling within each group.

Table S4. Source-label stratification of the subjective contrasts.

Source	Position	Item	Contrast	Difference [95% CI]	Pairs
AV	Ego	Atypicality	A-W	+0.333 [+0.008, +0.671]	20
AV	Ego	Atypicality	C-W	+0.842 [+0.512, +1.158]	20
AV	Ego	Atypicality	A-C	-0.508 [-0.787, -0.242]	20
AV	Ego	Q2 yielding	A-W	-0.129 [-0.604, +0.375]	20
AV	Ego	Q2 yielding	C-W	+0.887 [+0.533, +1.233]	20
AV	Ego	Q2 yielding	A-C	-1.017 [-1.363, -0.667]	20
AV	Ego	Q3 predictability	A-W	-0.717 [-0.971, -0.458]	20
AV	Ego	Q3 predictability	C-W	-0.654 [-0.850, -0.450]	20
AV	Ego	Q3 predictability	A-C	-0.062 [-0.287, +0.167]	20
AV	Ego	Comfort	A-W	-0.650 [-0.896, -0.438]	20
AV	Ego	Comfort	C-W	-0.354 [-0.546, -0.196]	20
AV	Ego	Comfort	A-C	-0.296 [-0.438, -0.138]	20
AV	CP	Atypicality	A-W	+0.571 [+0.129, +1.046]	20
AV	CP	Atypicality	C-W	+0.533 [+0.254, +0.825]	20
AV	CP	Atypicality	A-C	+0.037 [-0.242, +0.312]	20
AV	CP	Q2 yielding	A-W	-0.404 [-0.817, +0.025]	20
AV	CP	Q2 yielding	C-W	+0.508 [+0.250, +0.821]	20
AV	CP	Q2 yielding	A-C	-0.912 [-1.262, -0.558]	20
AV	CP	Q3 predictability	A-W	-0.867 [-1.071, -0.675]	20
AV	CP	Q3 predictability	C-W	-0.588 [-0.825, -0.362]	20
AV	CP	Q3 predictability	A-C	-0.279 [-0.542, +0.004]	20
AV	CP	Comfort	A-W	-1.017 [-1.229, -0.825]	20
AV	CP	Comfort	C-W	-0.242 [-0.408, -0.087]	20

Source	Position	Item	Contrast	Difference [95% CI]	Pairs
AV	CP	Comfort	A-C	-0.775 [-0.963, -0.588]	20
AV	CP	Reported cost	A-W	+2.046 [+1.800, +2.296]	20
AV	CP	Reported cost	C-W	+0.858 [+0.671, +1.054]	20
AV	CP	Reported cost	A-C	+1.188 [+0.938, +1.404]	20
AV	CP	Q5b naturalness	A-W	+0.083 [-0.083, +0.237]	20
AV	CP	Q5b naturalness	C-W	+0.204 [+0.058, +0.346]	20
AV	CP	Q5b naturalness	A-C	-0.121 [-0.304, +0.071]	20
HUMAN	Ego	Atypicality	A-W	+0.533 [+0.154, +0.887]	20
HUMAN	Ego	Atypicality	C-W	+0.600 [+0.383, +0.812]	20
HUMAN	Ego	Atypicality	A-C	-0.067 [-0.537, +0.400]	20
HUMAN	Ego	Q2 yielding	A-W	+0.208 [-0.171, +0.592]	20
HUMAN	Ego	Q2 yielding	C-W	+0.317 [+0.058, +0.571]	20
HUMAN	Ego	Q2 yielding	A-C	-0.108 [-0.371, +0.142]	20
HUMAN	Ego	Q3 predictability	A-W	-0.629 [-0.863, -0.412]	20
HUMAN	Ego	Q3 predictability	C-W	-0.346 [-0.546, -0.146]	20
HUMAN	Ego	Q3 predictability	A-C	-0.283 [-0.504, -0.067]	20
HUMAN	Ego	Comfort	A-W	-0.692 [-0.867, -0.517]	20
HUMAN	Ego	Comfort	C-W	-0.375 [-0.558, -0.188]	20
HUMAN	Ego	Comfort	A-C	-0.317 [-0.517, -0.129]	20
HUMAN	CP	Atypicality	A-W	+0.683 [+0.321, +1.062]	20
HUMAN	CP	Atypicality	C-W	+0.783 [+0.633, +0.925]	20
HUMAN	CP	Atypicality	A-C	-0.100 [-0.483, +0.283]	20
HUMAN	CP	Q2 yielding	A-W	+0.029 [-0.300, +0.396]	20
HUMAN	CP	Q2 yielding	C-W	+0.458 [+0.188, +0.763]	20
HUMAN	CP	Q2 yielding	A-C	-0.429 [-0.696, -0.163]	20
HUMAN	CP	Q3 predictability	A-W	-0.767 [-0.996, -0.546]	20
HUMAN	CP	Q3 predictability	C-W	-0.458 [-0.688, -0.221]	20
HUMAN	CP	Q3 predictability	A-C	-0.308 [-0.487, -0.133]	20
HUMAN	CP	Comfort	A-W	-1.154 [-1.288, -1.021]	20
HUMAN	CP	Comfort	C-W	-0.192 [-0.358, -0.013]	20
HUMAN	CP	Comfort	A-C	-0.963 [-1.113, -0.812]	20
HUMAN	CP	Reported cost	A-W	+2.108 [+1.900, +2.329]	20
HUMAN	CP	Reported cost	C-W	+0.975 [+0.804, +1.146]	20
HUMAN	CP	Reported cost	A-C	+1.133 [+0.946, +1.329]	20
HUMAN	CP	Q5b naturalness	A-W	-0.017 [-0.167, +0.125]	20
HUMAN	CP	Q5b naturalness	C-W	-0.025 [-0.188, +0.133]	20
HUMAN	CP	Q5b naturalness	A-C	+0.008 [-0.138, +0.158]	20

This sensitivity excludes records labelled `alt27=1` in the supplied coding and preserves the primary analysis separately. The table reports available participants and the existing pair-bootstrap intervals; no primary records are removed by this display.

Table S5. Sensitivity to excluding the supplied alternative-code records.

Position	Item	Contrast	Difference [95% CI]	Participants
Ego	Atypicality	A-W	+0.565 [+0.231, +0.903]	40
Ego	Atypicality	C-W	+0.745 [+0.605, +0.879]	40
Ego	Atypicality	A-C	-0.180 [-0.481, +0.117]	40
Ego	Q2 yielding	A-W	+0.022 [-0.324, +0.379]	40
Ego	Q2 yielding	C-W	+0.522 [+0.271, +0.769]	40
Ego	Q2 yielding	A-C	-0.500 [-0.745, -0.260]	40
Ego	Q3 predictability	A-W	-0.707 [-0.891, -0.528]	40
Ego	Q3 predictability	C-W	-0.505 [-0.675, -0.325]	40
Ego	Q3 predictability	A-C	-0.203 [-0.368, -0.038]	40
Ego	Comfort	A-W	-0.720 [-0.845, -0.598]	40
Ego	Comfort	C-W	-0.376 [-0.489, -0.265]	40
Ego	Comfort	A-C	-0.343 [-0.470, -0.223]	40
CP	Atypicality	A-W	+0.698 [+0.352, +1.080]	40
CP	Atypicality	C-W	+0.709 [+0.566, +0.855]	40
CP	Atypicality	A-C	-0.011 [-0.282, +0.272]	40
CP	Q2 yielding	A-W	-0.187 [-0.444, +0.083]	40
CP	Q2 yielding	C-W	+0.393 [+0.210, +0.584]	40
CP	Q2 yielding	A-C	-0.580 [-0.808, -0.355]	40
CP	Q3 predictability	A-W	-0.846 [-1.034, -0.662]	40
CP	Q3 predictability	C-W	-0.516 [-0.698, -0.340]	40
CP	Q3 predictability	A-C	-0.330 [-0.530, -0.130]	40
CP	Comfort	A-W	-1.111 [-1.244, -0.986]	40
CP	Comfort	C-W	-0.226 [-0.352, -0.106]	40
CP	Comfort	A-C	-0.885 [-0.987, -0.784]	40
CP	Reported cost	A-W	+2.083 [+1.885, +2.282]	40
CP	Reported cost	C-W	+0.950 [+0.802, +1.107]	40
CP	Reported cost	A-C	+1.132 [+0.979, +1.280]	40
CP	Q5b naturalness	A-W	+0.048 [-0.073, +0.163]	40
CP	Q5b naturalness	C-W	+0.075 [-0.035, +0.183]	40
CP	Q5b naturalness	A-C	-0.027 [-0.150, +0.097]	40

Each contrast gives equal weight to segments within scenario and then to shared

scenarios. Intervals resample shared scenarios 10,000 times with the observed raters fixed. This estimand differs from the participant-paired analysis and is not an independent replication.

Table S6. Equal-common-scenario sensitivity for all subjective endpoints.

Position	Item	Contrast	Difference [95% CI]	Scenarios
Ego	Atypicality	A-W	+0.665 [+0.098, +1.246]	8
Ego	Atypicality	C-W	+0.928 [+0.656, +1.206]	9
Ego	Atypicality	A-C	-0.263 [-0.643, +0.122]	12
Ego	Q2 yielding	A-W	+0.166 [-0.365, +0.685]	8
Ego	Q2 yielding	C-W	+0.581 [+0.140, +1.089]	9
Ego	Q2 yielding	A-C	-0.691 [-1.192, -0.127]	12
Ego	Q3 predictability	A-W	-0.644 [-0.736, -0.532]	8
Ego	Q3 predictability	C-W	-0.493 [-0.638, -0.380]	9
Ego	Q3 predictability	A-C	-0.178 [-0.341, -0.028]	12
Ego	Comfort	A-W	-0.650 [-0.783, -0.517]	8
Ego	Comfort	C-W	-0.352 [-0.424, -0.276]	9
Ego	Comfort	A-C	-0.323 [-0.400, -0.230]	12
CP	Atypicality	A-W	+0.850 [+0.144, +1.656]	8
CP	Atypicality	C-W	+0.949 [+0.542, +1.413]	9
CP	Atypicality	A-C	-0.072 [-0.480, +0.352]	12
CP	Q2 yielding	A-W	-0.300 [-0.674, +0.073]	8
CP	Q2 yielding	C-W	+0.368 [-0.091, +0.812]	9
CP	Q2 yielding	A-C	-0.759 [-1.249, -0.310]	12
CP	Q3 predictability	A-W	-0.775 [-0.952, -0.571]	8
CP	Q3 predictability	C-W	-0.485 [-0.649, -0.339]	9
CP	Q3 predictability	A-C	-0.337 [-0.502, -0.156]	12
CP	Comfort	A-W	-1.120 [-1.212, -1.020]	8
CP	Comfort	C-W	-0.245 [-0.417, -0.079]	9
CP	Comfort	A-C	-0.851 [-0.982, -0.725]	12
CP	Reported cost	A-W	+2.005 [+1.839, +2.179]	8
CP	Reported cost	C-W	+0.915 [+0.771, +1.068]	9
CP	Reported cost	A-C	+1.191 [+0.991, +1.403]	12
CP	Q5b naturalness	A-W	+0.044 [-0.079, +0.164]	8
CP	Q5b naturalness	C-W	+0.110 [-0.018, +0.247]	9
CP	Q5b naturalness	A-C	-0.087 [-0.227, +0.052]	12

Supplementary Note 4. Complete counterpart-response battery

The four existing counterpart-response endpoints below retain both directions, including heading and yaw endpoints not shown on the main speed-response axis. Ratios compare pooled moment-weighted medians. Pointwise 95% intervals use 2,000 whole-run bootstrap draws over 175 scenario runs. A–W is the frozen assertive battery; C–W is the previously reported post-hoc accommodating extension. The reference value is one; intervals containing one do not establish equivalence. No absolute-difference interval is inferred from these ratios.

Table S7. Complete counterpart-response median ratios.

Endpoint	Contrast	Ratio [95% CI]	Side moments	W moments
Speed reduction	A–W	+2.062 [+1.057, +3.873]	469	10483
Speed reduction	C–W	+0.790 [+0.279, +1.535]	719	10483
Speed range	A–W	+1.886 [+1.349, +2.378]	469	10483
Speed range	C–W	+1.076 [+0.782, +1.529]	719	10483
Net heading change	A–W	+2.351 [+0.905, +11.319]	469	10483
Net heading change	C–W	+0.890 [+0.107, +3.485]	719	10483
Peak yaw rate	A–W	+2.042 [+0.894, +8.353]	469	10483
Peak yaw rate	C–W	+0.849 [+0.199, +2.933]	719	10483

Extended Data

a The human reference range rises, falls and changes width with the situation

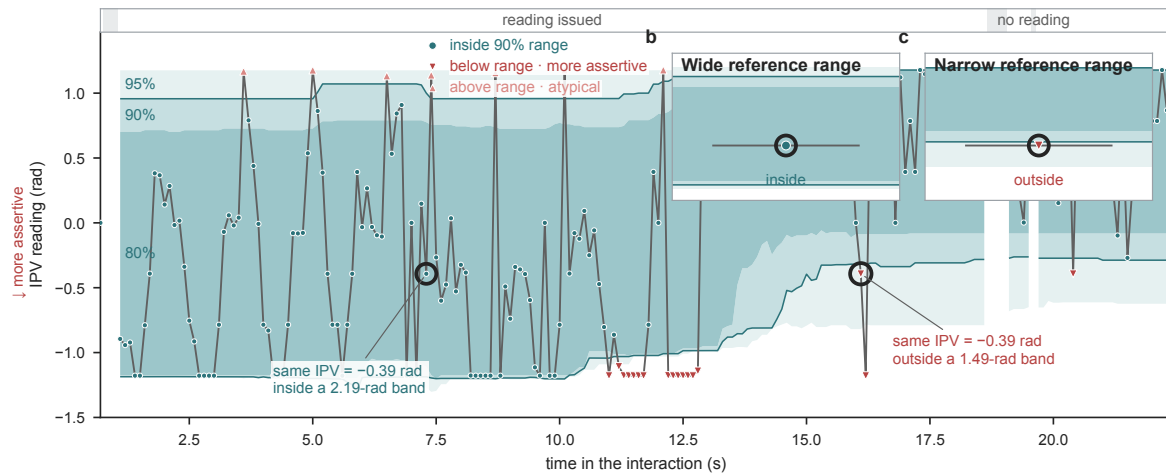


Figure ED1. The human reference range rises, falls and changes width with the situation. A single interaction segment showing the frame-level reading against the nested 80%, 90% and 95% situation-conditioned human reference ranges. Grey intervals in the top strip mark frames with no reading; those frames remain gaps rather than being plotted as a value. Readable frames are shown as inside the 90% range, below it (more assertive, atypical) or above it (the accommodating side, atypical). The two outlined frames carry the same reading, -0.393 rad: it lies inside the range where that range spans 2.19 rad and outside it where the range spans 1.49 rad. The pair isolates the role of conditioning, in that the number being judged is identical while the reference against which it is judged is not. The display is descriptive. Source data are provided as a Source Data file.

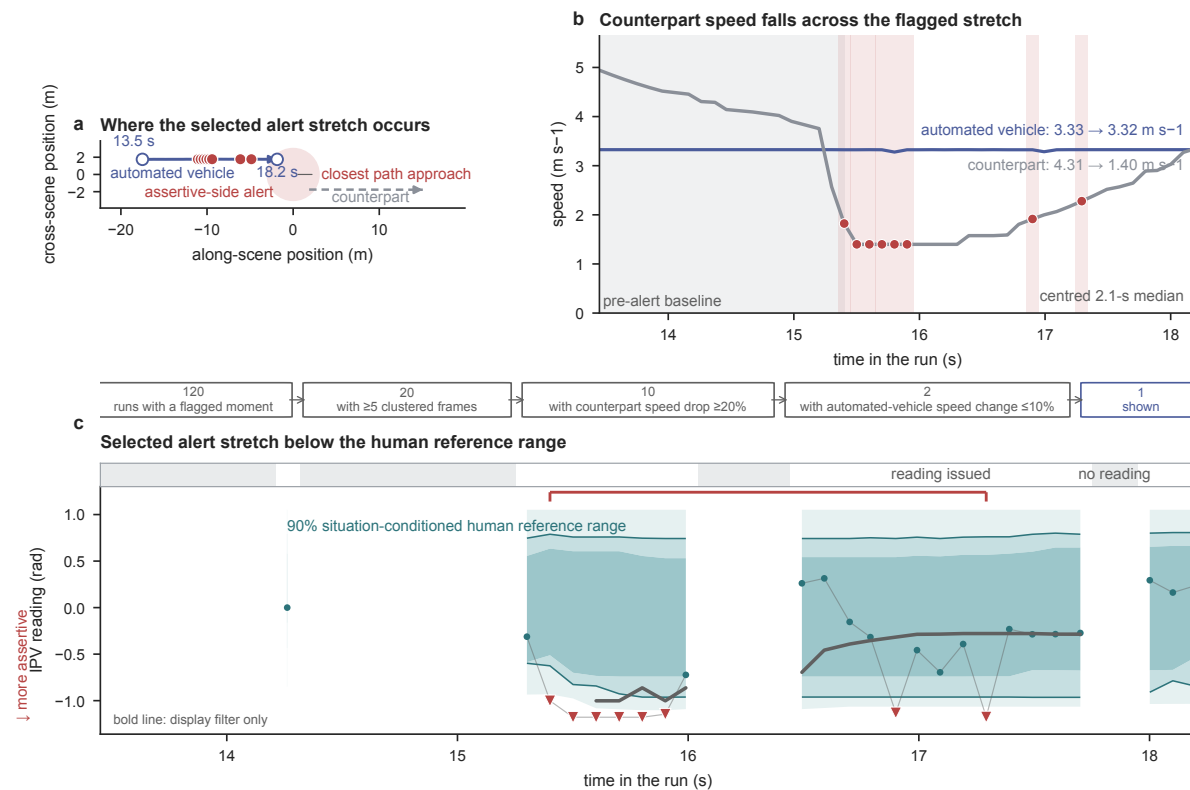


Figure ED2. One illustrative run containing flagged assertive moments. **a** Bird's-eye trajectories over the displayed window; red points mark the monitored vehicle's positions during the selected flagged stretch. **b** Speed timelines over a 2.0 s pre-flag baseline and the flagged stretch. Lines are centred 21-frame medians of speed computed from the logged velocity components. The counterpart median moves from 4.31 to 1.40 m s⁻¹; the automated vehicle's median moves from 3.33 to 3.32 m s⁻¹. **c** The same stretch relative to the 90% situation-conditioned human reference range. Two traces are drawn from the same data: the faint line and its markers are the per-frame readings, which are what the verdicts are computed on, and the bold line is those readings under a centred 2.1-second median—the filter the speed panel above also uses—which is what a display would show. Filtering is applied to the shown value only and never to a verdict; the bold line is masked to readable frames so it does not bridge an abstention. Red downward triangles are per-frame readings below the lower edge of the range; grey ticks mark no reading, and available rows are connected only for display continuity. This run was selected from the 120 runs containing at least one flagged assertive moment, and is one of only two meeting display criteria fixed in advance (at least five flagged frames in one cluster, successive flags separated by at most one second; counterpart speed decrease of at least 20%; automated-vehicle speed change within 10%). Of the two runs meeting all criteria, the displayed one ranks first under the selection score fixed together with them — the counterpart's speed-drop fraction minus the automated vehicle's absolute speed-change fraction (0.675 vs 0.552). It is one illustrative run: it is not a typical value, not an overall distribution, and not evidence for ranking or comparison between automated systems. The display is descriptive and assigns no causation. Source data are provided as a Source Data file.

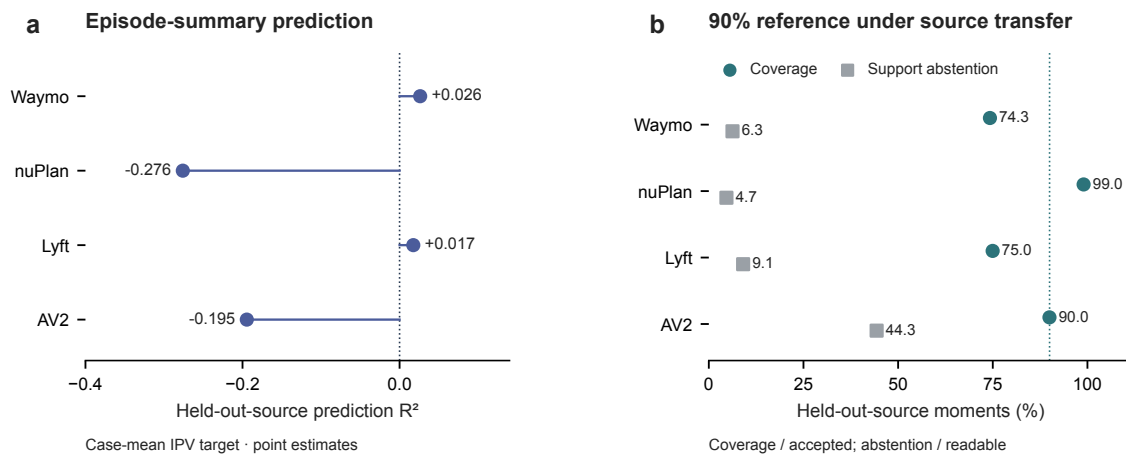


Figure ED3. Source transfer of the behavioural predictor and of the reference monitor are distinct checks. **a** Full-state-space prediction of event-mean IPV in each source omitted during fitting, retaining negative out-of-source R^2 values. Stems connect point estimates to zero and are not uncertainty intervals. **b** Empirical coverage of the 90% reference among accepted readable moments, shown alongside human-support abstention among all readable moments in each omitted source. Coverage and abstention have different denominators; both are needed to interpret transfer. The four sources remain separate and no universal transfer guarantee is inferred. These are existing source-held-out results, not new fits. Source Data contain exact evaluated, accepted and abstaining counts.

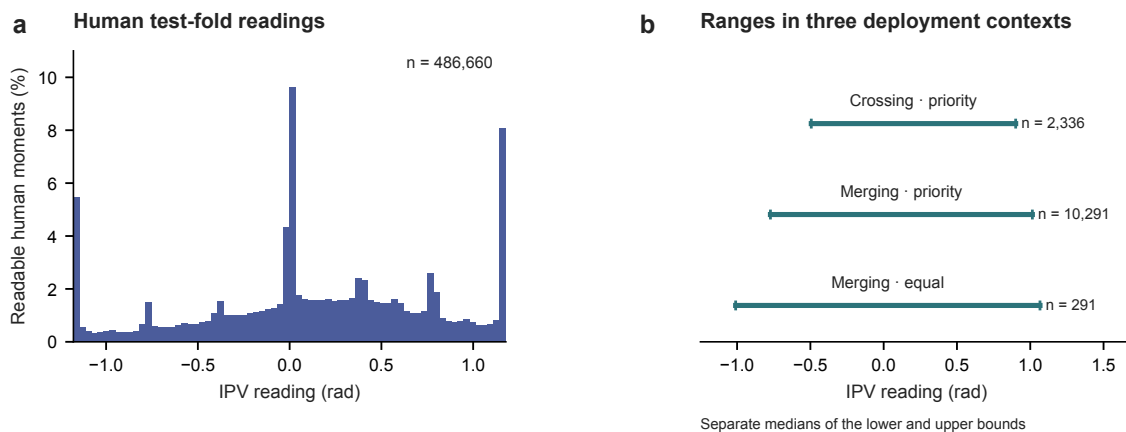


Figure ED4. The human reading distribution and deployment reference summaries describe different populations. **a** Frozen distribution of 486,660 readable natural-driving test moments. **b** Three illustrative deployment situations, summarised by the separate medians of the frozen 90% lower and upper bounds within each situation class. These summaries use all corresponding automated-benchmark candidate moments, before readability or human-support filtering: 2,336 crossing-with-priority, 10,291 merging-with-priority and 291 merging-with-equal-priority moments. The endpoints are marginal medians and need not form any individual moment's interval. Across all 67,861 benchmark candidates, predicted widths span 1.05–2.37 rad. These deployment summaries are not intervals calculated from the human histogram in **a**. Source Data preserve the two populations separately.

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Author Contributions

X.Z. conceived the study, developed the methodology and the software, and wrote the original draft and the revisions. J.S. acquired funding and revised the manuscript. X.J. contributed validation and data curation. Y.L. contributed methodology, data curation and validation. G.M. contributed to the conception of the study and revised the manuscript.

Competing Interests

The matched-scenario real-vehicle benchmark analysed in this work is organised by the authors' research group at Tongji University. No author entered the challenge as a competitor, the official outcome labels examined here are publicly released, and the funding for this study is independent of, and does not overlap with, the funding of the challenge series. The authors declare no other competing interests.